

Unit 6 — Waves, Sound & Light

*wave properties · $v = f\lambda$ · reflection · refraction · diffraction · interference · sound
· resonance · Doppler · EM spectrum · mirrors + lenses · RADAR · Wien's law ·
seismic + tsunami*

◆ NASA / PLANETARY DEFENSE BRIEFING — UPDATE 6

For five units, you have been computing things about 2026-XJ. Distance. Velocity. Mass. Trajectory. Impact yield. Damage radii. Every one of those numbers had to come from a measurement.

But HOW did NASA measure them? You haven't been told. The answer is RADAR.

For decades, NASA's planetary radar network has tracked near-Earth asteroids by sending out radio pulses and timing the return. The TIME tells distance. The frequency SHIFT tells velocity (Doppler effect). The signal STRENGTH tells size. Every number you've used was derived from bounced radio waves.

Unit 6 reveals the measurement chain. By Day 15, you will walk through the FULL chain — radar pulse out, radar echo back, what each piece of the signal tells us about 2026-XJ.

Two new pieces of data this unit:

(1) 2026-XJ has a 70% chance of impacting OCEAN, not land. Earth is 70% water. If it splashes down instead of landing, the damage mechanism shifts from thermal-pulse-and-shockwave to TSUNAMI. Day 17 builds the tsunami physics.

(2) Meteor color tells temperature. The atmospheric-entry glow from Day 10 of Unit 5 is yellow-white. Why not red, why not blue? Wien's law connects color directly to temperature. Day 16 explains.

This is fiction. The physics — and the equations — are real.

Name: _____	Class period: _____	Date
received: _____		

"Every number we have used about 2026-XJ came from bounced radio waves."

Table of Contents

Day-type key: A = Anchor · S = Standard · L = Lab · T = Transfer · ★ = Asteroid payoff

Section	Type	Page
Cover	—	1
Table of Contents	—	2
Phase 1 — Wave fundamentals	—	
Day 1 — Waves carry energy, not matter (re-anchor)	A	3
Day 2 — Wave anatomy: f , λ , A , T	S	6
Day 3 — The wave equation: $v = f\lambda$	S	9
Phase 2 — Wave behavior	—	
Day 4 — Reflection at boundaries	S	12
Day 5 — Refraction: bending at a boundary	S	15
Day 6 — Diffraction: bending around obstacles	S	18
Day 7 — Interference + standing waves	S	21
Phase 3 — Sound	—	
Day 8 — Sound as longitudinal wave	S	24
Day 9 — Speed of sound + resonance	S	27
Day 10 — Vernier echo lab	L	30
Day 11 — Intensity, decibels, Doppler	S	33
Phase 4 — Light + EM	—	
Day 12 — EM spectrum: light is a wave	S	36
Day 13 — Plane mirrors + virtual images	S	39
Day 14 — Lenses qualitatively	S	42
Phase 5 — Asteroid payoff	—	
Day 15 — ★ RADAR — the reveal	S★	45
Day 16 — Meteor color (Wien's law)	S★	48
Day 17 — Seismic + tsunami damage at distance	S★	51
Phase 6 — Assessment	—	
Day 18 — Unit 6 transfer task	T	54
Equation Reference Card	—	57
Mastery Growth Chart	—	59

Day 1 — Waves Carry Energy — Not Matter (Re-anchor + Unit Preview)

Today's Target

I can describe how a wave carries ENERGY and INFORMATION through a medium without carrying matter, and name the unit's three payoff days (Day 15 radar, Day 16 meteor color, Day 17 seismic + tsunامي).

ASTEROID THREAD

What we know about 2026-XJ today: From Units 1-5: 2026-XJ — distance, velocity, mass, impact yield, damage radii. But every NUMBER had to come from a measurement.

How today connects: The measurement was RADAR. Radio waves SENT, BOUNCED, RECEIVED — and Doppler-shifted by the moving asteroid. Unit 6 is the framework that makes radar work. Day 15 is the reveal.

NOTICE AND WONDER

Demo: a slinky stretched along the floor. Two students hold the ends. Pulse the slinky side-to-side (transverse). Then push-pull (longitudinal). Watch CAREFULLY. What TRAVELED down the slinky? What did NOT travel?

I notice _____

I wonder _____

DOODLE ZONE

Trace what happens to a single coil during each kind of pulse. Does any coil move all the way across the slinky? What DOES travel from end to end?

Demo: a slinky stretched along the floor. Two students hold ends.

Pulse side-to-side (transverse) — then push-pull (longitudinal).

A — TRANSVERSE PULSE (medium moves PERPENDICULAR to wave direction)



B — LONGITUDINAL PULSE (medium moves ALONG wave direction)



What TRAVELED down the slinky — coils? Or energy? _____

VOCAB · WAVE

Definition: A disturbance that transfers ENERGY (and information) through a medium without transferring matter. The medium oscillates in place; only the pattern moves.

Example: A floating cork bobs up and down as an ocean wave passes — but doesn't move with the wave. Sound makes air molecules wiggle in place; the COMPRESSION pattern travels.

Cognates: *Sp. onda · Pt. onda · HC vag*

Watch out: The medium does NOT travel. Only the DISTURBANCE travels. A wave can move kilometers; the water molecules barely shift.

VOCAB · TRANSVERSE WAVE

Definition: The medium oscillates PERPENDICULAR to the direction the wave travels.

Example: A rope shaken up and down. Light waves. Seismic S-waves.

Cognates: *Sp. onda transversal · Pt. onda transversal · HC vag transvèsal*

Watch out: Picture the wave moving →, but the medium wiggling ↑. Two different directions.

VOCAB · LONGITUDINAL WAVE

Definition: The medium oscillates PARALLEL to the direction the wave travels. Compressions and rarefactions.

Example: Sound waves in air. A slinky pushed-and-pulled along its length. Seismic P-waves.

Cognates: *Sp. onda longitudinal · Pt. onda longitudinal · HC vag longitudinal*

Watch out: A wave's TYPE (transverse / longitudinal) is about how the medium moves, NOT about what carries the wave.

MY GUESS — IF A WAVE CAN TRANSFER ENERGY WITHOUT TRANSFERRING MATTER, HOW COULD A SIGNAL FROM EARTH EVER MEASURE SOMETHING MILLIONS OF KM AWAY? (DAY 15 IS THE ANSWER.)



Unit 6 has THREE payoff days. Day 15: RADAR — every 2026-XJ number traces to bounced radio waves. Day 16: METEOR COLOR — Wien's law connects glow color to temperature. Day 17: SEISMIC + TSUNAMI — mechanical waves carry impact damage across the planet.

EXIT TICKET — DAY 1

You wave the end of a rope. A pulse travels down the rope. What's moving through the rope: rope material, energy, or both? Explain in two sentences.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can describe what travels in a wave (energy + info — not matter).	☐	☐	☐	☐
I can distinguish transverse vs. longitudinal.	☐	☐	☐	☐
I can name the three Unit 6 payoff days.	☐	☐	☐	☐

Day 2 — Wave Anatomy: Frequency · Wavelength · Amplitude · Period

Today's Target

I can identify f , λ , A , and T on a wave diagram, give correct units for each, and use $f = 1/T$ to convert between frequency and period.

ASTEROID THREAD

What we know about 2026-XJ today: Radar uses radio waves around $f \approx 10^9$ Hz (gigahertz). At $c = 3 \times 10^8$ m/s, that's $\lambda \approx 0.3$ m wavelength.

How today connects: Today: the vocabulary that lets us describe radar precisely. Day 3: the equation that connects them.

VOCAB · WAVELENGTH (λ)

Definition: Distance from one crest to the next crest (or any equivalent point). A SPACE measurement. Units: m, mm, nm.

Example: Visible light ≈ 500 nm. Sound (at 440 Hz, A4) ≈ 0.78 m. Ocean swell ≈ 30 m. Radio FM ≈ 3 m.

Cognates: *Sp. longitud de onda · Pt. comprimento de onda · HC longè vag*

Watch out: λ is in METERS, not seconds. It's a distance. The Greek letter looks like a wishbone.

VOCAB · AMPLITUDE (A)

Definition: Maximum displacement from the rest position. A SPACE measurement. Units: m for transverse, Pa for sound pressure.

Example: A loud sound has bigger A than a soft sound at the same pitch. Bright light has bigger A than dim light at the same color.

Cognates: *Sp. amplitud · Pt. amplitude · HC anplitid*

Watch out: A is the MAX displacement, not crest-to-trough. The full peak-to-peak is $2A$.

VOCAB • PERIOD (T)

Definition: Time for ONE complete cycle. A TIME measurement. Units: seconds (s).

Example: A 440 Hz tuning fork has $T = 1/440 \approx 0.00227$ s. Earth's orbit: $T = 1$ year.

Cognates: Sp. *período* · Pt. *período* · HC *perýdò*

Watch out: T is SECONDS PER CYCLE. f is CYCLES PER SECOND. They are reciprocals.

VOCAB • FREQUENCY (F)

Definition: Number of cycles per second. Units: hertz (Hz). $f = 1/T$.

Example: A 440 Hz tuning fork vibrates 440 times per second. FM radio at 95.7 MHz vibrates 9.57×10^7 times per second.

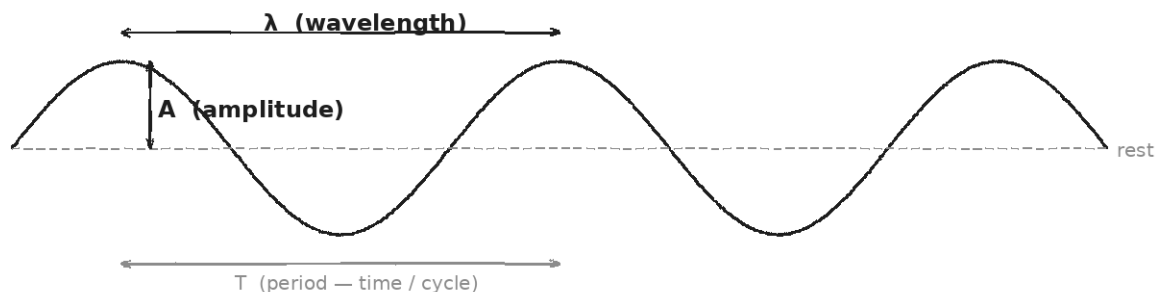
Cognates: Sp. *frecuencia* · Pt. *frequência* · HC *fwekans*

Watch out: Higher f = MORE cycles per second = shorter T. For sound: higher f = higher pitch.

DOODLE ZONE

Label each property — λ , A, T — on the sine wave with arrows. Then identify which label is a SPACE measurement and which is a TIME measurement.

Four words describe any wave: **WAVELENGTH (λ)** · **AMPLITUDE (A)** · **PERIOD (T)** · **FREQUENCY (f)**
 λ and A are space measurements; T and f are time measurements.



Reciprocal pair: $f = 1 / T$ $T = 1 / f$ Cycles per second = HERTZ (Hz).

WORKED EXAMPLE — CONVERT T TO F

A wave has period $T = 0.004$ s. What's its frequency?

GIVEN	$T = 0.004$ s
EQUATION	$f = 1 / T$
WORK	$f = 1 / 0.004$ $f = 250$ Hz
ANSWER	$f = 250$ Hz (250 cycles per second).

WORKED EXAMPLE — CONVERT F TO T

A tuning fork has $f = 440$ Hz. What's its period?

GIVEN	$f = 440$ Hz
EQUATION	$T = 1 / f$
WORK	$T = 1 / 440$ $T \approx 0.00227$ s $T \approx 2.27$ ms
ANSWER	$T \approx 2.27$ milliseconds.

YOUR TURN — PERIOD OF A RADAR PULSE

A radar transmitter operates at $f = 2.4 \text{ GHz} = 2.4 \times 10^9 \text{ Hz}$. What's its period (in seconds and nanoseconds)?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	ONE 2.4 GHz radio pulse takes about 0.42 nanoseconds per cycle. In a $1 \mu\text{s}$ pulse, that's 2400 cycles. Day 15 will show why pulses contain THOUSANDS of cycles — for precise distance + Doppler measurement.
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Fill in the frame:

f tells us _____ per second. T
tells us seconds per _____. They
are reciprocals, so high f means
_____ T .

EXIT TICKET — DAY 2

A wave has $f = 250$ Hz. What's its period? Show one line of work.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can label λ , A, T, f on a wave diagram.	☐	☐	☐	☐
I can convert between f and T using $f = 1/T$.	☐	☐	☐	☐
I can give correct units for each property.	☐	☐	☐	☐

Day 3 — The Wave Equation: $v = f \cdot \lambda$

Today's Target

I can apply $v = f\lambda$ to compute any one of v , f , or λ given the other two, and explain WHY a wave changing media keeps the same f but changes λ .

ASTEROID THREAD

What we know about 2026-XJ today: Radar pulses travel at $v = c = 3 \times 10^8$ m/s. The round-trip time tells distance.

How today connects: Today's equation IS the radar equation in disguise. Once you trust $v = f\lambda$ for everyday waves, you trust $d = c \cdot \Delta t / 2$ for radar.

VOCAB · $V = F \cdot \lambda$

Definition: Speed of a wave equals frequency times wavelength. Same equation for water, sound, light, radio.

Example: Sound at 440 Hz with $\lambda = 0.78$ m: $v = 440 \cdot 0.78 \approx 343$ m/s (sound in air).

Cognates: Sp. $v = f \cdot \lambda$ · Pt. $v = f \cdot \lambda$ (universal symbols)

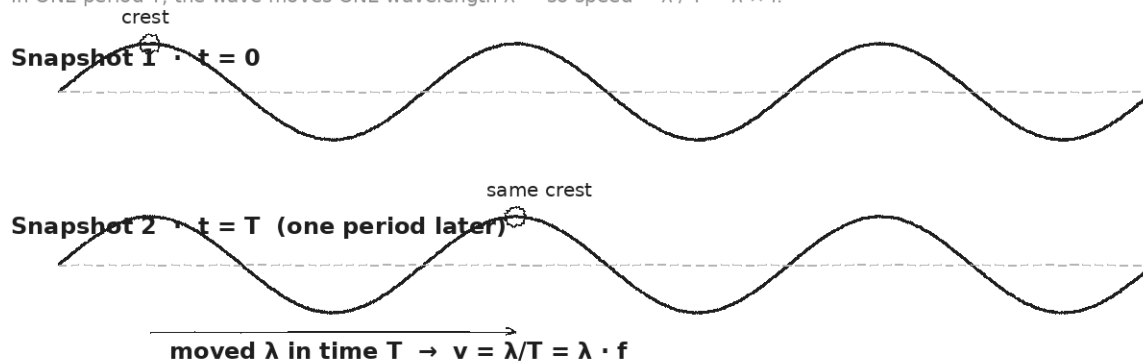
Watch out: v depends on the MEDIUM. f is set by the SOURCE. λ adjusts: $\lambda = v/f$. Same source, different medium $\rightarrow$ different λ .

DOODLE ZONE

Two snapshots of the same wave, one period apart. Mark a single crest in both. How far did it travel? Write the relationship: $\text{speed} = \text{distance} / \text{time}$.

$$v = f \cdot \lambda \quad (\text{speed of a wave})$$

In ONE period T , the wave moves ONE wavelength λ — so $\text{speed} = \lambda / T = \lambda \times f$.



WORKED EXAMPLE — SOUND'S WAVELENGTH IN AIR AT A4

An A4 tuning fork at 440 Hz makes sound that travels at $v = 343 \text{ m/s}$ in air. What's the wavelength?

GIVEN	$f = 440 \text{ Hz} \cdot v = 343 \text{ m/s}$
EQUATION	$v = f \cdot \lambda \rightarrow \lambda = v / f$
WORK	$\lambda = 343 / 440$ $\lambda \approx 0.78 \text{ m}$
ANSWER	$\lambda \approx 0.78 \text{ m} \approx 78 \text{ cm}.$

WORKED EXAMPLE — FM RADIO WAVELENGTH

An FM radio station broadcasts at $f = 100 \text{ MHz} = 10^8 \text{ Hz}$. Radio waves travel at $c = 3 \times 10^8 \text{ m/s}$. What's the wavelength?

GIVEN	$f = 10^8 \text{ Hz} \cdot c = 3 \times 10^8 \text{ m/s}$
EQUATION	$\lambda = c / f$
WORK	$\lambda = (3 \times 10^8) / 10^8$ $\lambda = 3 \text{ m}$
ANSWER	$\lambda = 3 \text{ m}$. An FM antenna sized at $\lambda/4 \approx 75 \text{ cm}$ captures these waves efficiently.

YOUR TURN — A4 IN WATER

The same A4 tuning fork ($f = 440 \text{ Hz}$) sounds underwater. Sound in water moves at $v \approx 1480 \text{ m/s}$. What's the new wavelength?

GIVEN	
EQUATION	
WORK	
ANSWER	

YOUR TURN — RADAR WAVELENGTH

NASA's planetary radar transmits at $f = 2.4 \text{ GHz} = 2.4 \times 10^9 \text{ Hz}$. Radio in vacuum travels at $c = 3 \times 10^8 \text{ m/s}$. What's λ ?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	Same source, two media. Same FREQUENCY (set by the source). But because v differs in each medium, the WAVELENGTH adjusts to fit. This is exactly why refraction (Day 5) happens at a boundary.
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Fill in the frame:

When a wave enters a new medium, the SOURCE doesn't change. So f stays the _____ . But v changes, so by $v = f\lambda$, the _____ must change. If v decreases, λ _____ .

EXIT TICKET — DAY 3

An FM radio wave has $f = 95$ MHz. What's its wavelength? (Use $c = 3 \times 10^8$ m/s.)

TODAY'S TARGETS — RATE YOURSELF

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Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can apply $v = f\lambda$ in any direction (solve for v , f , or λ).	☐	☐	☐	☐
I can explain why f is unchanged across media but λ changes.	☐	☐	☐	☐
I can compute λ for radio and sound at given frequencies.	☐	☐	☐	☐

Day 4 — Reflection at Boundaries: $\theta_i = \theta_r$ (from the normal)

Today's Target

I can apply the law of reflection to predict the reflected angle, measuring angles from the NORMAL — not from the surface — and explain why the law holds for sound, light, and radar.

ASTEROID THREAD

What we know about 2026-XJ today: Radar works because radio waves REFLECT off the asteroid. Without reflection, no echo, no distance, no Doppler.

How today connects: Today: the law that says EVERY wave reflects with equal angles. Day 15 applies this to the radar echo from 2026-XJ.

VOCAB · LAW OF REFLECTION

Definition: The angle of incidence equals the angle of reflection. Both angles are measured from the NORMAL (perpendicular to the surface).

Example: Light at 35° from normal reflects at 35° from normal — on the OTHER side of the normal. Same for radar off an asteroid.

Cognates: *Sp. ley de reflexión · Pt. lei da reflexão · HC lwa refleksyon*

Watch out: Measure from the NORMAL, not the surface. A common mistake: " 35° from the mirror" — that's actually 55° from the normal.

VOCAB · NORMAL

Definition: A line perpendicular to the surface at the point where the wave hits.

Example: On a flat tabletop, the normal points straight up. On a curved mirror, the normal is perpendicular to the tangent at that point.

Cognates: *Sp. normal · Pt. normal · HC nòmál*

Watch out: "Normal" in physics means perpendicular — not "usual".

VOCAB · TRANSMISSION (VS. REFLECTION)

Definition: When a wave hits a boundary, SOME reflects and SOME transmits into the new medium. The fraction depends on the impedance contrast between the two media.

Example: A wave hits a brick wall → almost all reflects. Hits a curtain → much passes through. Light through window → most transmits.

Cognates: Sp. *transmisión* · Pt. *transmissão* · HC *transmisyon*

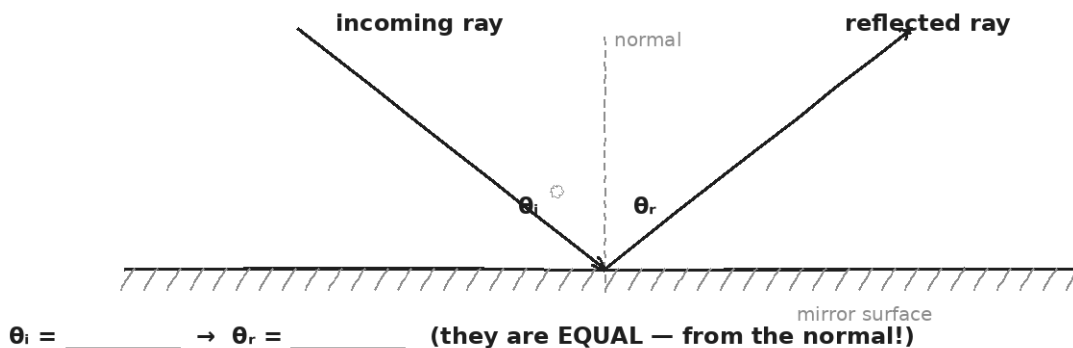
Watch out: Reflection + transmission together conserve the wave's energy (minus any absorbed).

DOODLE ZONE

Mark angle of incidence (θ_i) and angle of reflection (θ_r) — both from the NORMAL, not from the mirror surface. Verify the angles are equal.

LAW OF REFLECTION · angle of incidence = angle of reflection

Both angles measured FROM THE NORMAL (perpendicular to surface).



WORKED EXAMPLE — LASER OFF A MIRROR

You shine a laser at a mirror so the beam makes a 35° angle with the NORMAL. At what angle does it reflect?

GIVEN	$\theta_i = 35^\circ$ (from normal)
EQUATION	$\theta_r = \theta_i$
WORK	$\theta_r = 35^\circ$
ANSWER	The beam reflects at 35° from the normal, on the OTHER side of the normal.

WORKED EXAMPLE — ANGLE GIVEN FROM SURFACE

A radar beam strikes a flat asteroid surface at 20° from the SURFACE (i.e., a glancing hit). At what angle from the surface does it reflect?

GIVEN	20° from surface $\rightarrow 70^\circ$ from normal
EQUATION	θ_r (from normal) = θ_i (from normal) = 70°
WORK	θ_r (from surface) = $90^\circ - 70^\circ = 20^\circ$
ANSWER	20° from the surface — same as it came in. (Always convert to angle-from-normal first.)

YOUR TURN — SOUND OFF A WALL

A sound wave hits a wall at 50° from the normal. At what angle does the echo return?
What if measured from the wall surface?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	A perfectly mirrored asteroid would reflect the radar straight back if hit head-on. Real asteroids have ROUGH surfaces — radar scatters in many directions. The echo we detect is whatever reflects toward Earth. Signal strength $\times$ distance ² $\approx$ size estimate (Day 15).
---	--

Fill in the frame:

When a wave hits a flat surface, the angle measured from the
_____ on the way IN equals the
angle on the way OUT. We never measure from the
_____ itself.

EXIT TICKET — DAY 4

You shine a laser at a mirror at 35° from the normal. At what angle does it reflect? If you accidentally measured 35° from the mirror surface, what would the reflected angle be in that frame?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can apply the law of reflection from the normal.	☐	☐	☐	☐
I can convert between angle-from-normal and angle-from-surface.	☐	☐	☐	☐
I can explain WHY radar works (waves reflect).	☐	☐	☐	☐

Day 5 — Refraction: Speed Change → Direction Change

Today's Target

I can predict whether a wave bends TOWARD or AWAY from the normal as it enters a new medium, given which medium is faster.

ASTEROID THREAD

What we know about 2026-XJ today: Radio waves don't refract much in vacuum, but they DO refract in Earth's atmosphere. Sound from a distant impact refracts in air (sound bends toward cooler air).

How today connects: The pencil-in-water effect IS refraction. Same rule explains why distant thunder can be heard around hills.

VOCAB · REFRACTION

Definition: The BENDING of a wave at a boundary, caused by a change in wave SPEED. Slower medium → bends TOWARD normal. Faster medium → bends AWAY.

Example: Light going from air (faster) into water (slower) bends TOWARD the normal. That's why a pencil in a glass of water looks broken at the surface.

Cognates: *Sp. refracción · Pt. refração · HC refraksyon*

Watch out: Refraction needs a SPEED CHANGE, not just a boundary. If two media have the same wave speed, no refraction occurs.

VOCAB · INDEX OF REFRACTION (N) — ADVANCED

Definition: A ratio: $n = c / v_{\text{in_medium}}$. Higher n = slower wave. Air ≈ 1.00 , water ≈ 1.33 , glass ≈ 1.50 .

Example: In glass ($n = 1.5$), light travels at 2×10^8 m/s — two-thirds of c .

Cognates: *Sp. índice de refracción · Pt. índice de refração*

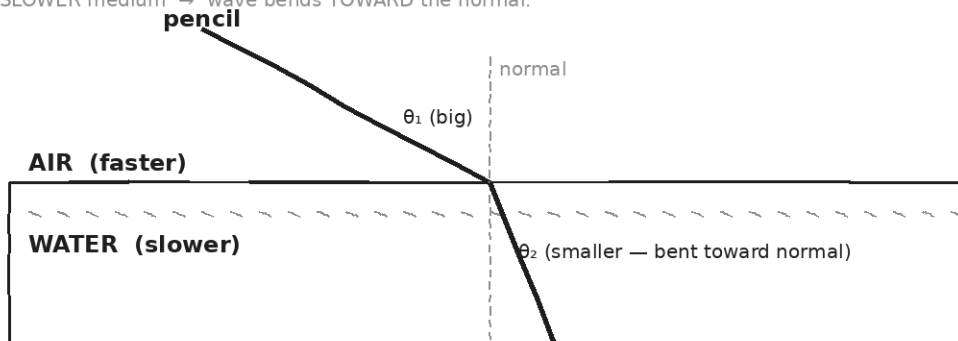
Watch out: Big n = SLOWER medium. The bigger the n , the more the wave slows and the more it bends toward normal.

DOODLE ZONE

The pencil enters water at an angle. Draw the normal at the air-water boundary. Which way does the wave bend? Mark θ_1 (above) and θ_2 (below).

REFRACTION · speed change → direction change

Entering a SLOWER medium → wave bends TOWARD the normal.



Bigger speed change → bigger bend. Why is the underwater part of the pencil tilted different

WORKED EXAMPLE — PREDICT THE BEND DIRECTION

Light enters water from air. Does it bend TOWARD or AWAY from the normal? Why?

GIVEN	Light in air: $v \approx c$ (faster) · Light in water: $v \approx 0.75c$ (slower)
EQUATION	Slower medium → bends TOWARD normal
WORK	Water is the slower medium. Therefore the light bends TOWARD the normal in water. θ_2 (in water) $<$ θ_1 (in air).
ANSWER	Bends TOWARD normal. The refracted ray is closer to vertical than the incident ray.

WORKED EXAMPLE — COMING BACK OUT

A swimmer underwater shines a flashlight UP through the water surface into the air. Does the beam bend toward or away from the normal as it exits water into air?

GIVEN	Light leaves water (slow) and enters air (fast).
EQUATION	Faster medium → bends AWAY from normal
WORK	Air is the faster medium. Therefore the light bends AWAY from the normal as it leaves water.
ANSWER	AWAY from the normal. The beam tilts further from vertical as it exits.

YOUR TURN — RADIO IN ATMOSPHERE

Radio waves enter Earth's atmosphere from space. The ionosphere slows them slightly compared to vacuum. Which way would the waves bend?

GIVEN	
EQUATION	
WORK	
ANSWER	

	Marching-band analogy: a row of soldiers marching at angle into mud. The side hitting mud first slows; the row twists. Same with waves entering the slower medium. The geometry follows.
---	--

Fill in the frame:

When a wave enters a SLOWER medium, the side that hits the boundary first _____ first. That makes the wavefront _____, bending it _____ the normal.

EXIT TICKET — DAY 5

Light enters glass ($n = 1.5$) from air ($n = 1.0$). Does it bend toward or away from the normal? In one sentence, explain why.

TODAY'S TARGETS — RATE YOURSELF

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Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can predict bend direction from speed change.	☐	☐	☐	☐
I can apply the rule to light + sound + radar.	☐	☐	☐	☐
I can explain refraction with the marching-band analogy.	☐	☐	☐	☐

Day 6 — Diffraction: Waves Bend Around Obstacles + Through Openings

Today's Target

I can predict how much diffraction occurs given the ratio of wavelength to obstacle size, and explain why we can HEAR around a corner but not SEE around one.

ASTEROID THREAD

What we know about 2026-XJ today: Long-wavelength radar diffracts around mountains (over-the-horizon radar). Short-wavelength radar doesn't diffract around small objects — better RESOLUTION for asteroid imaging.

How today connects: Engineers CHOOSE the radar wavelength based on what they want to detect. Diffraction sets the trade-off.

VOCAB · DIFFRACTION

Definition: The spreading of a wave as it passes by an obstacle or through an opening. More diffraction when the opening is comparable to or smaller than the wavelength.

Example: Sound ($\lambda \approx 1$ m) diffracts easily around doorways. Visible light ($\lambda \approx 500$ nm) barely diffracts around everyday objects — that's why we get sharp shadows.

Cognates: *Sp. difracción · Pt. difração · HC difraksyon*

Watch out: More diffraction when $\lambda \approx$ obstacle size. If obstacle is much BIGGER than λ , the wave just passes straight (light through a window) or stops (light by a wall).

VOCAB · RESOLUTION (IN RADAR / IMAGING) — PREVIEW

Definition: The minimum size of detail you can distinguish. Limited by diffraction — you can't resolve features smaller than $\sim\lambda$.

Example: Optical telescope ($\lambda \approx 500$ nm) can see details down to nm-scale on a microscope. Radar at $\lambda \approx 0.1$ m can resolve asteroid features down to ~ 0.1 m.

Cognates: *Sp. resolución · Pt. resolução · HC rezolisyon*

Watch out: Shorter wavelength = better resolution. That's why X-ray crystallography ($\lambda \approx 0.1$ nm) can image molecules.

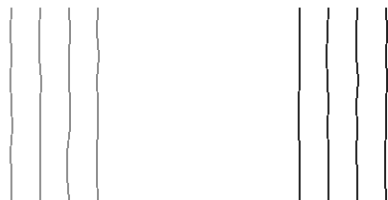
DOODLE ZONE

Compare the wide slit vs. narrow slit. Which produces a wave-fan? Why? What sets the amount of fanning — the slit size, the wavelength, or BOTH?

DIFFRACTION · waves bend around obstacles + through openings

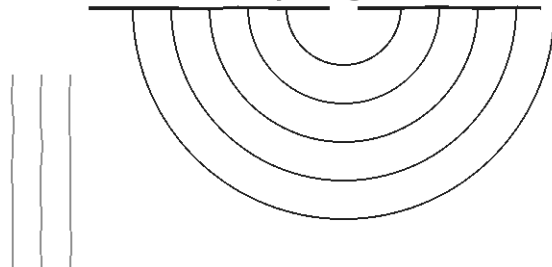
Smaller opening (compared to λ) → MORE diffraction.

A — WIDE SLIT (opening $\gg \lambda$)



→ wave passes mostly straight through

B — NARROW SLIT (opening $\approx \lambda$)



→ wave SPREADS in a fan (strong diffraction)

WORKED EXAMPLE — WHY YOU HEAR BUT DON'T SEE AROUND A CORNER

Sound has $\lambda \approx 1$ m. Visible light has $\lambda \approx 500$ nm = 5×10^{-7} m. A doorway is 1 m wide. Compare each wave's λ -to-opening ratio. Which diffracts strongly through the doorway?

GIVEN	$\lambda_{\text{sound}} = 1$ m · $\lambda_{\text{light}} = 5 \times 10^{-7}$ m · opening width = 1 m
EQUATION	ratio = λ / opening → compare
WORK	$\lambda_{\text{sound}} / 1$ m = 1 (comparable) $\lambda_{\text{light}} / 1$ m = 5×10^{-7} (tiny) Sound: ratio ≈ 1 , strong diffraction. Light: ratio $\approx 10^{-7}$, essentially no diffraction.
ANSWER	Sound diffracts → you HEAR around the doorway. Light doesn't diffract → you can't SEE around the doorway.

YOUR TURN — PICK A RADAR WAVELENGTH

You want to image an asteroid the size of a city block (~100 m across) but resolve features ~1 m in size. Should you use long-wavelength radio (~10 m) or short-wavelength radar (~0.1 m)? Explain.

GIVEN	
EQUATION	
WORK	
ANSWER	



Radio telescopes can see HUGE features in galaxies that optical telescopes can't — long wavelengths diffract around the dust clouds blocking visible light. Different wavelengths = different VIEWS of the universe.

Fill in the frame:

Diffraction happens when a wave passes through an opening or around an obstacle whose size is comparable to its

_____. The bigger the ratio of λ to obstacle size, the MORE the wave

_____.

EXIT TICKET — DAY 6

Why can you hear a conversation around a corner but not see the speaker? Answer with wavelengths.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can predict diffraction strength from λ vs. obstacle size.	☐	☐	☐	☐
I can explain hear-vs-see-around-corner using wavelengths.	☐	☐	☐	☐
I can explain the radar wavelength trade-off.	☐	☐	☐	☐

Day 7 — Interference + Standing Waves: When Two Waves Meet, They ADD

Today's Target

I can predict constructive vs. destructive interference based on whether two waves are in phase, and identify a standing wave as the interference of a wave with its reflection.

ASTEROID THREAD

What we know about 2026-XJ today: Modern radar uses INTERFEROMETRY — combining signals from multiple receivers to resolve fine asteroid details much smaller than a single antenna could see.

How today connects: Day 9 builds on standing waves to explain how guitar strings + organ pipes (and Earth's seismic resonances on Day 17) work.

VOCAB · INTERFERENCE

Definition: When two waves meet, their displacements ADD at each point. Same-direction crests → bigger crest (CONSTRUCTIVE). Opposite displacements → cancellation (DESTRUCTIVE).

Example: Two stones dropped in a pond — ripples meet and create bright + dark patterns. Noise-canceling headphones produce a sound exactly opposite to incoming noise — they CANCEL.

Cognates: *Sp. interferencia · Pt. interferência · HC entèfewans*

Watch out: Interference doesn't destroy energy. In destructive zones, the energy is just shifted to constructive zones elsewhere.

VOCAB · CONSTRUCTIVE INTERFERENCE

Definition: Two waves IN PHASE: crests align with crests, troughs with troughs. Net displacement = sum (bigger).

Example: Two speakers playing the same tone reinforce each other at certain spots = LOUD.

Cognates: *Sp. interferencia constructiva · Pt. interferência construtiva*

Watch out: In phase doesn't mean identical — it means crests align in TIME and SPACE.

VOCAB · DESTRUCTIVE INTERFERENCE

Definition: Two waves OUT OF PHASE by half a wavelength: crests align with troughs. Net displacement = 0 (cancellation).

Example: Noise-canceling headphones detect incoming noise and play the EXACT OPPOSITE — they cancel at your eardrum.

Cognates: *Sp. interferencia destructiva · Pt. interferência destrutiva*

Watch out: Energy isn't destroyed — it's redistributed. At dark fringes the energy is zero; at bright fringes it's doubled.

VOCAB · STANDING WAVE

Definition: A wave interfering with its own reflection. Produces fixed NODES (no motion) and ANTINODES (max motion). The wave LOOKS stationary.

Example: A guitar string fixed at both ends. Pluck it — a standing wave forms. The fundamental has nodes at the ends and one antinode in the middle.

Cognates: *Sp. onda estacionaria · Pt. onda estacionária · HC vag estasyonè*

Watch out: A standing wave is REAL motion of the medium — the wave just doesn't TRANSLATE left-right. It oscillates up-down at fixed locations.

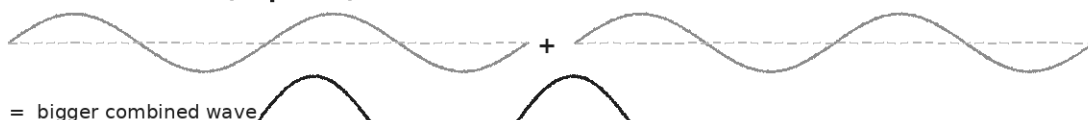
DOODLE ZONE

TOP: two waves *IN PHASE* — sum. BOTTOM: two waves *OUT OF PHASE* — sum. Mark the amplitudes (in phase = double, out of phase = zero).

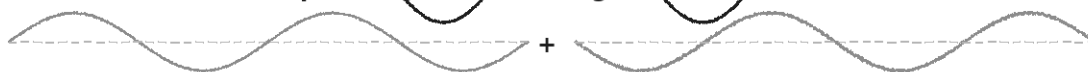
INTERFERENCE • two waves meet and ADD

Crest + Crest = BIGGER. Crest + Trough = ZERO (destructive).

A — CONSTRUCTIVE (in phase)



B — DESTRUCTIVE (out of phase by half wavelength)



= flat line (waves CANCEL)

Noise-canceling headphones generate the **OPPOSITE** wave — destructive!

WORKED EXAMPLE — NOISE-CANCELING HEADPHONES

Incoming engine noise enters as a 200 Hz wave with amplitude $A = 1$ (relative units). The headphones detect it and produce a 200 Hz wave with amplitude $A = 1$, but **SHIFTED** by half a wavelength. What's the net amplitude at your eardrum?

GIVEN	Both waves: 200 Hz, $A = 1$. Headphone wave shifted by $\lambda/2$ (opposite phase).
EQUATION	$A_{\text{net}} = A_1 + A_2$ (where A_2 is OPPOSITE-signed)
WORK	At every instant: $A_{\text{net}} = (+1) + (-1) = 0$ Continuously for both waves $\rightarrow$ silence at the eardrum.
ANSWER	$A_{\text{net}} = 0 \rightarrow$ silence. (Headphones never PERFECTLY cancel — but ~ 30 dB attenuation is real.)

WORKED EXAMPLE — GUITAR STRING FUNDAMENTAL

A guitar string is 0.65 m long. The wave speed along the string is 80 m/s. What's the fundamental (lowest) standing-wave frequency?

GIVEN	$L = 0.65 \text{ m} \cdot v = 80 \text{ m/s} \cdot n = 1 \text{ (fundamental)}$
EQUATION	$f_1 = v / (2L)$
WORK	$f_1 = 80 / (2 \cdot 0.65)$ $f_1 = 80 / 1.30$ $f_1 \approx 61.5 \text{ Hz}$
ANSWER	$f_1 \approx 61.5 \text{ Hz}$. The second harmonic ($n = 2$) would be 123 Hz, third 184.5, etc.

YOUR TURN — PIPE ORGAN

A 1.0-m-long pipe organ tube, open at both ends, supports standing waves. Sound in air is 343 m/s. What's the fundamental frequency? (Use $f_1 = v/(2L)$.)

GIVEN	
EQUATION	
WORK	
ANSWER	



The frequencies a guitar string + pipe organ + bell + Earth's crust + even an asteroid PRODUCE are set by their geometry — which standing waves fit. That's why a long string has a low fundamental and a short string a high one.

Fill in the frame:

Two waves in phase produce

_____ interference. Two waves

out of phase by $\lambda/2$ produce

_____ interference. A standing

wave is what happens when a wave interferes with its own

_____.

EXIT TICKET — DAY 7

Noise-canceling headphones produce a sound wave OPPOSITE to incoming noise. Why does this cancel? Where does the noise energy go? Answer in two sentences.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can predict constructive vs. destructive interference.	☐	☐	☐	☐
I can identify a standing wave and locate nodes + antinodes.	☐	☐	☐	☐
I can compute $f_n = nv/(2L)$ for a fixed-end string.	☐	☐	☐	☐

Day 8 — Sound is a Longitudinal Wave · Pitch = f · Loudness = A

Today's Target

I can describe sound as a longitudinal mechanical wave, identify pitch with frequency and loudness with amplitude, and explain why sound needs a MEDIUM (no sound in space).

ASTEROID THREAD

What we know about 2026-XJ today: An asteroid impact generates HUGE atmospheric shockwaves and INFRASOUND (< 20 Hz). The infrasound from the 2013 Chelyabinsk impact traveled around the entire planet — twice.

How today connects: Today: the foundation. Day 17 applies sound + seismic-wave physics to predict the damage and detection range from an asteroid strike.

VOCAB · SOUND (AS A WAVE)

Definition: A longitudinal mechanical wave — compressions and rarefactions in a medium (air, water, solid). Cannot travel through vacuum.

Example: A tuning fork pushes air molecules forward (compression), then pulls them back (rarefaction). The pattern travels outward at $v \approx 343$ m/s in air.

Cognates: *Sp. sonido · Pt. som · HC son*

Watch out: Sound is NOT made of moving air. Air molecules wiggle in place; only the COMPRESSION pattern moves. No medium = no sound.

VOCAB · PITCH

Definition: Our perception of a sound's frequency. Higher f = higher pitch. Units: Hz.

Example: Middle C ≈ 262 Hz. A4 = 440 Hz. Soprano high note ≈ 1500 Hz. Bass drum ≈ 50 Hz.

Cognates: *Sp. tono · Pt. altura · HC ton*

Watch out: Pitch is the PERCEPTUAL name; frequency is the physical quantity. Same idea, but pitch is what your brain hears.

VOCAB · LOUDNESS

Definition: Our perception of a sound's amplitude (or intensity). Bigger A = louder. Often measured in decibels on a logarithmic scale.

Example: A whisper has small A; a rock concert has large A. Doubling the AMPLITUDE more than doubles loudness perception.

Cognates: *Sp. intensidad · Pt. intensidade · HC fòs*

Watch out: Loudness is logarithmic in our ears. +10 dB feels about twice as loud, even though it's $\times 10$ intensity.

VOCAB · INFRASOUND + ULTRASOUND

Definition: INFRASOUND is sound below 20 Hz (we can't hear it). ULTRASOUND is above 20 kHz (we can't hear it). Both are real sound waves; just outside our range.

Example: Asteroid impacts produce powerful infrasound. Bats use ultrasound for echolocation. Medical ultrasound imaging operates near 1 MHz.

Cognates: *Sp. infrasonido / ultrasonido · Pt. infrassom / ultrassom*

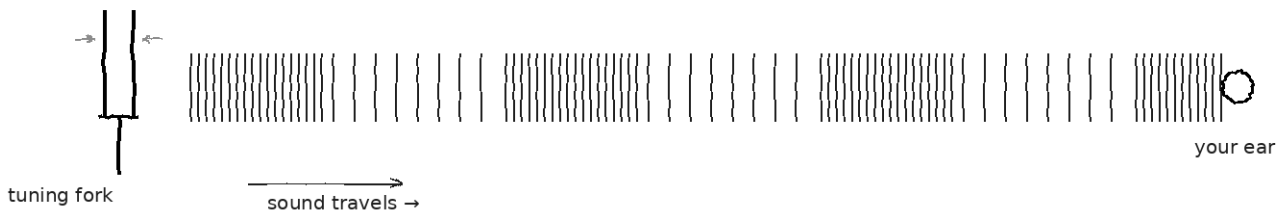
Watch out: Infra = below; ultra = above. Both are bands we don't hear, but other animals (and instruments) do.

DOODLE ZONE

Pattern shows alternating dense + sparse vertical lines. Label one COMPRESSION and one RAREFACTION. Mark the direction the SOUND travels — is it the same as the molecule motion, or perpendicular?

SOUND = a longitudinal wave (compressions + rarefactions in air)

Higher f = higher pitch. Bigger A = louder. Needs a medium.



Compression = molecules pushed CLOSER. Rarefaction = pushed APART. Air molecules wiggle

WORKED EXAMPLE — WHY NO SOUND IN SPACE

In a sci-fi movie, an explosion in space is shown with a loud BOOM. From a physics perspective, what's wrong?

GIVEN	Sound = compression wave needing a medium
EQUATION	(qualitative — no equation needed)
WORK	Space between objects is nearly vacuum. Sound needs molecules to compress + rarefy. No molecules → no compressions → no sound wave.
ANSWER	The boom can't propagate. Realistic space explosions are SILENT.

WORKED EXAMPLE — SAME SOURCE, TWO MEDIA

A 1000 Hz tuning fork is held just above the surface of a lake. What are the WAVELENGTHS of the sound (a) in air ($v = 343$ m/s) and (b) in water ($v = 1480$ m/s)?

GIVEN	$f = 1000 \text{ Hz}$ · $v_{\text{air}} = 343$ · $v_{\text{water}} = 1480$
EQUATION	$\lambda = v / f$
WORK	(a) $\lambda_{\text{air}} = 343 / 1000 = 0.343 \text{ m}$ (b) $\lambda_{\text{water}} = 1480 / 1000 = 1.48 \text{ m}$ Same f , but λ in water is 4.3× larger.
ANSWER	Air: 0.343 m. Water: 1.48 m. Same f , faster medium → longer λ .

YOUR TURN — PERIOD OF A TUNING FORK

A 512 Hz tuning fork is struck. What's the period of one cycle? (Use $T = 1/f$.)

GIVEN	
EQUATION	
WORK	
ANSWER	

★	Infrasound is the asteroid impact's long-distance signature. Below 20 Hz, sound waves travel thousands of km with little attenuation. NOAA's infrasound monitoring network has detected every major asteroid airburst since 2000.
---	---

Fill in the frame:

Sound is a _____ wave

(compressions + rarefactions). Pitch is the perception of

_____. Loudness is the

perception of _____. Without a

medium, sound cannot _____.

EXIT TICKET — DAY 8

Why is there no sound in outer space, even from a violent event? Answer in two sentences using the word 'medium'.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can describe sound as a longitudinal wave.	☐	☐	☐	☐
I can connect frequency to pitch + amplitude to loudness.	☐	☐	☐	☐
I can explain why sound needs a medium.	☐	☐	☐	☐

Day 9 — Speed of Sound + Resonance — Why Instruments Sound the Way They Do

Today's Target

I can predict the speed of sound in different media (air, water, solid) and apply $f_n = nv/(2L)$ to find the fundamental + harmonics of a standing wave instrument.

ASTEROID THREAD

What we know about 2026-XJ today: Earth has natural resonant frequencies — when a major impact (or earthquake) happens, the planet 'rings' at characteristic frequencies.

How today connects: Day 17's seismic waves are launched at these resonances. A 2026-XJ impact would excite them measurably from the other side of the planet.

VOCAB · SPEED OF SOUND (V_SOUND)

Definition: The speed at which compression waves travel through a medium. Faster in stiff media, slower in compressible ones. Increases with temperature in gases.

Example: Air at 20 °C: ≈ 343 m/s. Water: ≈ 1480 m/s. Steel: ≈ 5960 m/s. Adding 1 °C to air adds about 0.6 m/s.

Cognates: *Sp. velocidad del sonido · Pt. velocidade do som · HC vitès son*

Watch out: A stiff medium (steel) transmits sound FASTER, not slower. Stiff = fast restoring force = fast wave.

VOCAB · RESONANCE

Definition: When you drive a system at one of its natural standing-wave frequencies, energy builds up. The system 'sings' loudly at that frequency.

Example: Push a kid on a swing at exactly the swing's natural period → they go very high. Hit the right note near a wine glass → it shatters. Earth rings at low-Hz frequencies after large earthquakes.

Cognates: *Sp. resonancia · Pt. ressonância · HC rezonans*

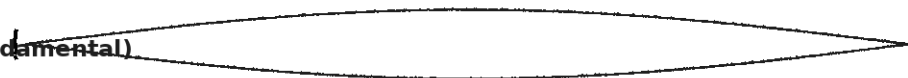
Watch out: Resonance requires MATCHING the natural frequency. Wrong frequency → energy doesn't build up.

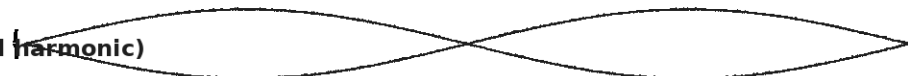
DOODLE ZONE

Three rows show $n = 1, 2, 3$ standing waves on a fixed-end string. For $n = 1$, $\lambda = \underline{\hspace{1cm}}$. Compute f_1 using $v = 80 \text{ m/s}$, $L = 0.65 \text{ m}$.

STANDING WAVE on a string · fundamental + harmonics

Both ends fixed. Only certain wavelengths fit. $f_n = n \cdot v / (2L)$.

$n = 1$ (fundamental)  $\lambda = 2L$

$n = 2$ (2nd harmonic)  $\lambda = L$

$n = 3$ (3rd harmonic)  $\lambda = 2L/3$

$L = 0.65 \text{ m}$, $v = 80 \text{ m/s} \rightarrow f_1 = v/(2L) = \underline{\hspace{1cm}} \text{ Hz}$ (your turn)

WORKED EXAMPLE — GUITAR STRING FUNDAMENTAL

A guitar string is 0.65 m long. Wave speed on the string is 80 m/s. What's the fundamental frequency?

GIVEN	$L = 0.65 \text{ m} \cdot v = 80 \text{ m/s} \cdot n = 1$
EQUATION	$f_n = n \cdot v / (2L)$
WORK	$f_1 = 1 \cdot 80 / (2 \cdot 0.65)$ $f_1 = 80 / 1.30$ $f_1 \approx 61.5 \text{ Hz}$
ANSWER	$f_1 \approx 61.5 \text{ Hz}$ (close to a low B on a bass guitar).

WORKED EXAMPLE — ECHO DISTANCE FROM 3-SECOND DELAY

You shout into a canyon and hear the echo 3.0 s later. How far away is the canyon wall? ($v_{\text{sound in air}} = 343 \text{ m/s}$.)

GIVEN	$\Delta t_{\text{round-trip}} = 3.0 \text{ s} \cdot v = 343 \text{ m/s}$
EQUATION	$d = v \cdot \Delta t / 2$ (round-trip $\rightarrow$ one-way)
WORK	$d = 343 \cdot 3.0 / 2$ $d = 1029 / 2$ $d \approx 514 \text{ m}$
ANSWER	$\approx 514 \text{ m}$. (Quick rule: 1 second of echo delay $\approx 170 \text{ m}$ to wall.)

YOUR TURN — HARMONICS OF AN OPEN PIPE

A 0.85-m pipe organ tube, open at both ends, has $v_{\text{sound}} = 343 \text{ m/s}$. Compute f_1 and f_2 .

GIVEN	
EQUATION	
WORK	
ANSWER	

★	The 1960 Chilean earthquake (9.5 magnitude) rang Earth like a bell for weeks. Some of its longest-period oscillations had $T \approx 54$ minutes — below 1 mHz infrasound. Day 17 returns to seismic resonance.
---	---

Fill in the frame:

A standing wave's frequency depends on the medium speed

_____ and the geometry

_____ — specifically, $f_1 = v/(2L)$.

The harmonics are multiples of

_____.

EXIT TICKET — DAY 9

A guitar string is 65 cm long. The wave speed on the string is 80 m/s. What's the fundamental frequency? Show one line of work.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? *Check the box if you want to talk with me about this target — even if you marked yourself a 3.*

Target	1	2	3	Help?
I can apply $f_n = nv/(2L)$ to a fixed-end string.	☐	☐	☐	☐
I can predict sound speed in different media.	☐	☐	☐	☐
I can compute echo distance from round-trip time.	☐	☐	☐	☐

Day 10 — Vernier Echo Lab · Measuring v_{sound} by Timing the Echo

Today's Target

I can use a Vernier microphone + LabQuest to time the delay between a clap and its echo from a known distance, compute $v_{\text{sound}} = 2L/\Delta t$, and compare to the accepted value with one credible source of error noted.

ASTEROID THREAD

What we know about 2026-XJ today: We've stated $v_{\text{sound}} \approx 343 \text{ m/s}$ in air. Today we measure it ourselves.

How today connects: If we trust $v = f\lambda$ + echo timing for everyday sound, we trust the same logic for radar pulse + Doppler analysis on Day 15.

LAB SETUP — INVESTIGATION 6.1

Driving question: Can we measure the speed of sound in our classroom by clapping toward a wall and timing the echo with a Vernier microphone?

Equipment:

- Vernier LabQuest + microphone (one per pair)
- Tape measure or laser distance finder
- Wall (or large flat surface) at least 10 m away
- Thermometer (room temp affects v_{sound})
- Spreadsheet or LoggerPro for waveform analysis

What you do:

1. Choose a wall and measure L (one-way distance) to ± 0.05 m. Record.
2. Set LabQuest sample rate to at least 10 kHz so the clap waveform is clean.
3. Have one partner clap sharply near the microphone. The other partner triggers data capture.
4. On the captured waveform, identify the clap peak + the echo peak. Read off Δt between them.
5. Compute $v_{\text{sound}} = 2L / \Delta t$. Round-trip distance is $2L$.
6. Repeat for 3 trials. Average. Note room temperature.
7. Compare to expected 343 m/s + temperature correction (about +0.6 m/s per $^{\circ}\text{C}$ above 20 $^{\circ}\text{C}$).

Safety: Loud claps near the microphone can damage hearing if amplified — keep volume low and don't clap directly at someone's ear.

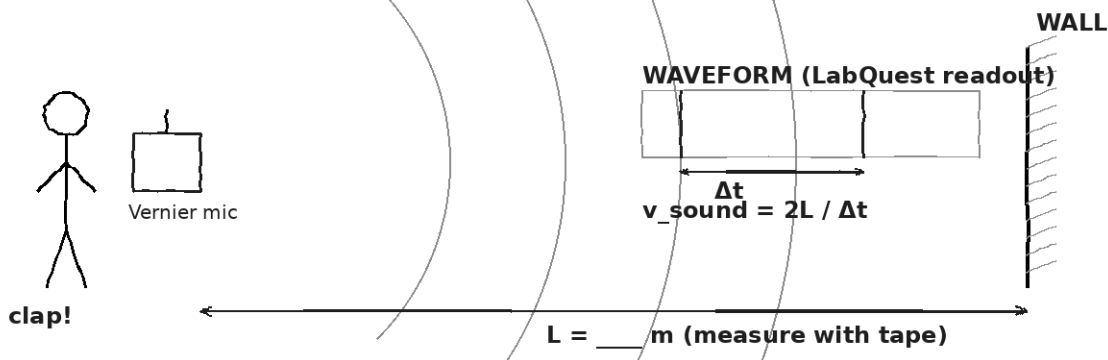
How this lab serves the year's question: If our v_{sound} is accurate to within 2% from a simple clap, NASA's Doppler radar (timing nanosecond pulses) achieves accuracy thousands of times better. Same physics, scaled.

DOODLE ZONE

Setup: clapper + Vernier mic + wall + tape measure. Label L . The waveform shows two peaks; mark Δt between them. Compute $v_{\text{sound}} = 2L/\Delta t$.

LAB SETUP • Measuring v_{sound} by echo timing

Clap near a wall. Mic records both the clap and the echo. Round trip = $2L$.



DATA TABLE — CLAP-ECHO TIMING

Trial	L (m)	Δt (s)	v_{sound} (m/s)	Notes
1				
2				
3				
4				
Mean v_{sound} :				

OBSERVATION & ANALYSIS

How close did your average come to 343 m/s? What's the percent difference?
Was the room warmer or cooler than 20 °C?

If you could only improve ONE of (L precision, Δt precision, room temperature control) — which would shrink your error the most?

Asteroid connection: how does this serve the year's question?



A 10 m wall + a 0.06 s round trip means Δt is small. Use a sample rate ≥ 10 kHz, or use a longer corridor (15-20 m) to spread the peaks apart.

Fill in the frame:

We measured $v_{\text{sound}} =$ _____

m/s. The accepted value at room temperature is

_____ m/s. Our percent

difference is _____ %.

The main source of error was likely _____.

EXIT TICKET — DAY 10

If you measured $v_{\text{sound}} = 350 \text{ m/s}$ and your wall was 10.0 m away, what Δt did you record? Show one line of work.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can run the echo-timing procedure correctly.	☐	☐	☐	☐
I can compute v_{sound} from $L + \Delta t$.	☐	☐	☐	☐
I can identify the largest source of error.	☐	☐	☐	☐

Day 11 — Intensity · Decibels · Doppler Effect

Today's Target

I can apply $I \propto 1/r^2$ for a point source, convert intensity to decibels using $\text{dB} = 10 \cdot \log_{10}(I/I_0)$, and predict whether a Doppler-shifted frequency rises or falls based on relative motion.

ASTEROID THREAD

What we know about 2026-XJ today: Doppler is HOW radar measures asteroid velocity. The echo's frequency shift from the transmit frequency tells the line-of-sight speed.

How today connects: Today: Doppler on a familiar scale (ambulance siren). Day 15: Doppler applied to 2026-XJ's radar return.

VOCAB · INTENSITY (I)

Definition: Power per unit area. Units: W/m^2 . Drops as $1/r^2$ for a point source (energy spread over a $4\pi r^2$ sphere).

Example: A 1 W bulb 1 m away delivers $I = 1/(4\pi) \approx 0.080 \text{ W/m}^2$. At 2 m, only $\frac{1}{4}$ as much (0.020 W/m^2).

Cognates: *Sp. intensidad · Pt. intensidade · HC entansite*

Watch out: Double the distance $\rightarrow \frac{1}{4}$ the intensity, not $\frac{1}{2}$. Inverse SQUARE law.

VOCAB · DECIBEL (DB)

Definition: A logarithmic measure of intensity. $\text{dB} = 10 \cdot \log_{10}(I / I_0)$, with $I_0 = 10^{-12} \text{ W/m}^2$ (threshold of hearing). Every +10 dB = $\times 10$ in intensity.

Example: 0 dB hearing threshold. 60 dB normal conversation. 120 dB pain. A rock concert (110 dB) is 10^{11} times more intense than the threshold.

Cognates: *Sp. decibelio · Pt. decibel · HC desibèl*

Watch out: dB is LOG scale. +20 dB $\neq$ +2 $\times$ intensity — it's +100 $\times$ intensity. Hearing damage starts around 85 dB with prolonged exposure.

VOCAB · DOPPLER EFFECT

Definition: When a wave SOURCE moves relative to an observer, the observed frequency SHIFTS. Approach → higher f . Recede → lower f .

Example: An ambulance siren goes from high pitch to low pitch as it passes you. Train horn rises in pitch as it approaches.

Cognates: *Sp. efecto Doppler · Pt. efeito Doppler · HC efè Doppler*

Watch out: Doppler is about RELATIVE motion. The SOURCE moving toward you gives the same shift as YOU moving toward the source.

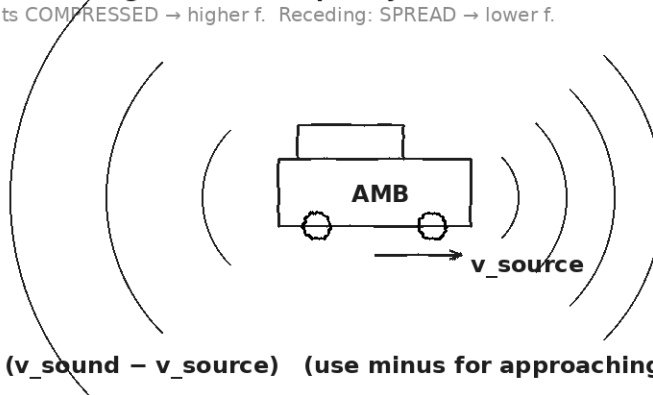
DOODLE ZONE

An ambulance moves right. Wavefronts AHEAD of it are compressed (higher f for the approaching listener). Wavefronts BEHIND it are stretched (lower f for the receder). Label which side hears HIGHER pitch.

DOPPLER EFFECT · moving source → frequency shift

Approaching: wavefronts COMPRESSED → higher f . Receding: SPREAD → lower f .

Receder
HEARS:
LOWER pitch



Approacher
HEARS:
HIGHER pitch

$$f' = f \cdot v_{\text{sound}} / (v_{\text{sound}} - v_{\text{source}}) \quad (\text{use minus for approaching, plus for receding})$$

WORKED EXAMPLE — DOPPLER SHIFT, AMBULANCE APPROACHING

An ambulance siren emits at $f = 800$ Hz. The ambulance approaches you at $v_{\text{source}} = 30$ m/s. Sound in air: $v = 343$ m/s. What frequency do you hear?

GIVEN	$f = 800$ Hz · $v = 343$ m/s · $v_{\text{source}} = 30$ m/s (toward listener)
EQUATION	$f' = f \cdot v / (v - v_{\text{source}})$
WORK	$f' = 800 \cdot 343 / (343 - 30)$ $f' = 800 \cdot 343 / 313$ $f' = 274\,400 / 313$ $f' \approx 877$ Hz
ANSWER	You hear ≈ 877 Hz — about a half-step higher in pitch.

WORKED EXAMPLE — DECIBEL RATIO

A whisper measures 30 dB. A motorcycle measures 100 dB. The motorcycle is how many times more INTENSE than the whisper?

GIVEN	$\Delta\text{dB} = 100 - 30 = 70$ dB
EQUATION	$\text{ratio} = 10^{(\Delta\text{dB} / 10)}$
WORK	$\text{ratio} = 10^{(70 / 10)}$ $\text{ratio} = 10^7$ $\text{ratio} = 10\,000\,000$
ANSWER	The motorcycle is 10 million times more INTENSE than the whisper. (Perceived loudness rises slower than intensity.)

YOUR TURN — TRAIN HORN RECEDING

A train at 600 Hz horn moves AWAY from you at 20 m/s. What frequency do you hear?
($v_{\text{sound}} = 343 \text{ m/s}$. Use plus sign for receding: $f' = f \cdot v / (v + v_{\text{source}})$.)

GIVEN	
EQUATION	
WORK	
ANSWER	

★	Same Doppler logic, scaled to radar: 2026-XJ at v_{source} returns radar with a tiny frequency shift Δf . NASA detects shifts as small as 0.01 Hz on a 2.4 GHz carrier. That's 4 parts in 10^{12} . Day 15 walks through the full chain.
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Fill in the frame:

A source moving TOWARD you compresses wavefronts → higher

_____ . Moving AWAY stretches

them → lower _____ . The dB

scale is _____ in intensity: every

+10 dB is _____ × more intense.

EXIT TICKET — DAY 11

A train at 600 Hz horn moves at 20 m/s. You stand on the platform as it APPROACHES. What frequency do you hear? (Use $v_{\text{sound}} = 343 \text{ m/s}$.) Show your work.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can apply $f' = f \cdot v / (v \mp v_{\text{source}})$ for approach vs. recession.	☐	☐	☐	☐
I can interpret the dB scale (logarithmic).	☐	☐	☐	☐
I can connect Doppler to radar velocity measurement.	☐	☐	☐	☐

Day 12 — Electromagnetic Spectrum · Light Is a Wave

Today's Target

I can identify the bands of the EM spectrum in order (radio → gamma), explain that all travel at c in vacuum, and compute λ from f using $c = f\lambda$ for any band.

ASTEROID THREAD

What we know about 2026-XJ today: Radar (radio band) and the meteor's atmospheric-entry GLOW (visible) are the SAME kind of wave family. Different f , same physics.

How today connects: Day 16 makes this concrete — the meteor's GLOW color (visible light) tells us its temperature (Wien's law).

VOCAB · ELECTROMAGNETIC WAVE

Definition: A transverse wave of oscillating electric + magnetic fields. Travels at $c = 3 \times 10^8$ m/s in vacuum. No medium needed.

Example: Light. Radio. X-rays. Microwaves. All the same kind of wave, different frequencies.

Cognates: *Sp. onda electromagnética · Pt. onda eletromagnética · HC onn elektwomayetik*

Watch out: EM waves are NOT mechanical — they need no medium. That's why they cross vacuum from the Sun to Earth.

VOCAB · EM SPECTRUM

Definition: The full range of EM frequencies, from low- f radio (km wavelengths) up to high- f gamma rays (sub-nm wavelengths).

Example: Radio (km), microwave (cm), infrared (μm), VISIBLE (400-700 nm), UV (10s of nm), X-ray (nm), gamma (sub-nm).

Cognates: *Sp. espectro electromagnético · Pt. espectro eletromagnético*

Watch out: The bands shade into each other — there's no sharp edge between, say, radio and microwave. The names are conventions.

VOCAB · VISIBLE LIGHT

Definition: The narrow band of EM the human eye detects, from ~400 nm (violet) to ~700 nm (red). The Sun emits its peak intensity here.

Example: ROYGBIV. Red has the longest visible λ ; violet the shortest. Color = wavelength.

Cognates: *Sp. luz visible · Pt. luz visível · HC limyè vizib*

Watch out: "Visible" is just our species' window. Other animals see UV (bees), IR (some snakes), etc.

DOODLE ZONE

Walk left-to-right through the EM bands. Which band has the LONGEST wavelength? The HIGHEST frequency? Where does our radar fall? Where does visible light fall?

ELECTROMAGNETIC SPECTRUM · all travel at $c = 3 \times 10^8$ m/s in vacuum

Same wave family. Different frequency $\leftrightarrow$ different wavelength.

RADIO	MICROWAVE	INFRARED	VISIBLE	UV	X-RAY	GAMMA
10^4 m	10^{-2} m	10^{-5} m	500 nm	10^{-8} m	10^{-10} m	$<10^{-12}$ m



VISIBLE LIGHT · ROYGBIV · ~400 nm (violet) to ~700 nm (red)

R 700	O 620	Y 580	G 530	B 470	I 440	V 400
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WORKED EXAMPLE — MICROWAVE WAVELENGTH

Your microwave oven operates at $f = 2.4 \text{ GHz} = 2.4 \times 10^9 \text{ Hz}$. What's its wavelength?
Which EM band is this?

GIVEN	$f = 2.4 \times 10^9 \text{ Hz} \cdot c = 3 \times 10^8 \text{ m/s}$
EQUATION	$\lambda = c / f$
WORK	$\lambda = (3 \times 10^8) / (2.4 \times 10^9)$ $\lambda = 0.125 \text{ m}$ $\lambda = 12.5 \text{ cm}$
ANSWER	$\lambda \approx 12.5 \text{ cm}$. This is in the MICROWAVE band — sized to make food molecules absorb.

WORKED EXAMPLE — VISIBLE GREEN LIGHT

Green light has $\lambda \approx 530 \text{ nm}$. What's its frequency?

GIVEN	$\lambda = 530 \text{ nm} = 5.30 \times 10^{-7} \text{ m} \cdot c = 3 \times 10^8 \text{ m/s}$
EQUATION	$f = c / \lambda$
WORK	$f = (3 \times 10^8) / (5.30 \times 10^{-7})$ $f = 5.66 \times 10^{14} \text{ Hz}$
ANSWER	$f \approx 5.66 \times 10^{14} \text{ Hz} \approx 566 \text{ THz}$ — about a half-quadrillion cycles per second.

YOUR TURN — X-RAY FREQUENCY

An X-ray has $\lambda \approx 1.0 \times 10^{-10}$ m. What's its frequency?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	Why is X-ray imaging dangerous but visible-light imaging safe? Energy per photon depends on frequency. Higher f = higher-energy photons. X-rays can ionize molecules (break bonds). Visible photons can't.
---	--

Fill in the frame:

All EM waves travel at _____ in vacuum. The EM spectrum is ordered from longest wavelength (_____) to shortest (_____). Visible light is just one narrow band, from $\sim$ _____ nm to $\sim$ _____ nm.

EXIT TICKET — DAY 12

What's the wavelength of a microwave oven's 2.4 GHz radiation? (Use $c = 3 \times 10^8$.)
Which EM band is this?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can identify EM spectrum bands in order.	☐	☐	☐	☐
I can apply $c = f\lambda$ across any band.	☐	☐	☐	☐
I can connect color of visible light to wavelength.	☐	☐	☐	☐

Day 13 — Plane Mirrors · Virtual Images Behind the Mirror

Today's Target

I can construct a ray diagram for a plane mirror, locate the virtual image at equal distance behind the mirror, and explain why the image is upright + same size.

ASTEROID THREAD

What we know about 2026-XJ today: Radio telescopes are giant CURVED mirrors that focus radio waves to a detector. The Day 4 reflection law applies directly.

How today connects: Today the FLAT-mirror case. Day 14 generalizes to curves (lenses). Day 15 mentions parabolic radio dishes that combine both ideas.

VOCAB · PLANE MIRROR

Definition: A flat reflecting surface. Light obeys the law of reflection (Day 4). Produces VIRTUAL images behind the mirror at equal distance from the object.

Example: Your bathroom mirror. A glass storefront. A still pond surface.

Cognates: *Sp. espejo plano · Pt. espelho plano · HC miwa plat*

Watch out: "Plane" means flat — not like an airplane. Curved mirrors (concave, convex) behave differently and are not on the CPA scope.

VOCAB · VIRTUAL IMAGE

Definition: An image that LIGHT DOES NOT ACTUALLY REACH. The brain TRACES light rays backward and locates the image at the convergence point. Cannot be projected on a screen.

Example: Your mirror image — the light reflects off the mirror and into your eye. The image APPEARS to be behind the mirror, but no light is actually there.

Cognates: *Sp. imagen virtual · Pt. imagem virtual · HC imaj vityèl*

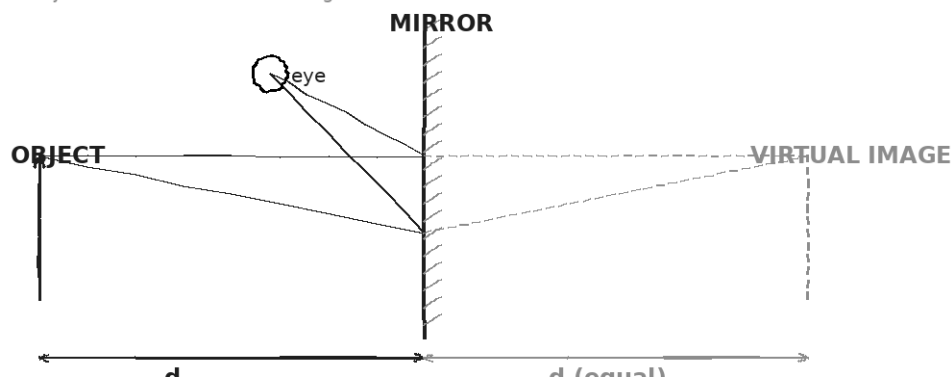
Watch out: A virtual image cannot be 'caught' on a screen. A real image CAN. Plane mirrors always make virtual images.

DOODLE ZONE

Object on the left of the mirror. Draw two rays from the object's tip — through the law of reflection. Trace each reflected ray BACKWARD (dashed). They converge at the VIRTUAL image, equal distance behind the mirror.

PLANE MIRROR · virtual image, same size, behind the mirror at equal distance

Pick two rays from the object — trace each one through the law of reflection.



WORKED EXAMPLE — IMAGE LOCATION

A 2.0-m-tall person stands 1.5 m from a flat mirror. Where is their image? How tall does it appear?

GIVEN	height = 2.0 m · d_object = 1.5 m from mirror
EQUATION	$d_{\text{image}} = d_{\text{object}}$ · $h_{\text{image}} = h_{\text{object}}$ (plane mirror)
WORK	$d_{\text{image}} = 1.5$ m behind the mirror. $h_{\text{image}} = 2.0$ m tall. Image is upright + virtual + same size.
ANSWER	1.5 m behind the mirror, 2.0 m tall, upright.

WORKED EXAMPLE — CLOSING THE DISTANCE

You walk 0.5 m closer to a mirror. By how much do you and your image close on each other?

GIVEN	You move 0.5 m toward mirror. Mirror is stationary.
EQUATION	$d_{\text{you}} = -0.5 \text{ m} \rightarrow d_{\text{image}} = +0.5 \text{ m}$ closer also (mirror)
WORK	Your distance to mirror: $\downarrow 0.5 \text{ m}$ Image's distance to mirror (= your distance): $\downarrow 0.5 \text{ m}$ Image moves 0.5 m TOWARD mirror also. Total closing: 1.0 m.
ANSWER	You and your image close on each other by 1.0 m total.

YOUR TURN — SHORTEST MIRROR TO SEE YOURSELF

What's the shortest mirror that lets you see your full body (top of head to feet)? Hint: trace two rays — one from your eyes to your feet, one from your eyes to the top of your head. The mirror only needs to cover the midpoints.

GIVEN	
EQUATION	
WORK	
ANSWER	



The answer to the YOUR TURN problem: a mirror HALF your height, mounted at the right height (top edge at eye level + half of forehead, bottom edge at feet midpoint), shows you head-to-toe regardless of how far away you stand.

Fill in the frame:

A plane mirror produces a _____
image (light doesn't actually reach it). The image is the same
_____ as the object, located the
same _____ behind the mirror as
the object is in front. It is _____
(upright/inverted).

EXIT TICKET — DAY 13

A 2-m-tall person stands 1.5 m from a flat mirror. Where is their image? How tall does it appear? Answer in one line.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can construct a ray diagram for a plane mirror.	☐	☐	☐	☐
I can identify virtual vs. real images.	☐	☐	☐	☐
I can predict image location + size for a plane mirror.	☐	☐	☐	☐

Day 14 — Lenses Qualitatively · Convex Converges · Concave Diverges

Today's Target

I can distinguish convex (converging) from concave (diverging) lenses by their shape and ray behavior, and identify whether the image is REAL or VIRTUAL.

ASTEROID THREAD

What we know about 2026-XJ today: Optical telescopes combine MIRRORS (objective) + LENSES (eyepiece) to magnify distant objects. Asteroid optical telescopes are limited by diffraction (Day 6) and lens / mirror size.

How today connects: Day 15 will mention that NASA's Goldstone radar dish is a parabolic reflector — a giant curved mirror for radio.

VOCAB · CONVEX LENS

Definition: A lens that is THICKER in the middle than at the edges. Refraction at the curved surfaces makes parallel rays CONVERGE to a focal point F at distance f from the lens.

Example: A magnifying glass. A camera lens. The eye's lens.

Cognates: *Sp. lente convexa · Pt. lente convexa · HC lant konvèks*

Watch out: "Convex" = bulging outward. "Converging" = brings rays together. Both apply.

VOCAB · CONCAVE LENS

Definition: A lens THINNER in the middle. Refraction makes parallel rays DIVERGE outward; tracing them back gives a virtual focal point on the incoming side.

Example: Eyeglasses for nearsightedness (myopia). A peephole. A door viewer (wide-angle).

Cognates: *Sp. lente cóncava · Pt. lente côncava · HC lant konkav*

Watch out: Concave lenses ALWAYS make virtual images — they spread rays out, can't focus them onto a screen.

VOCAB · FOCAL LENGTH (F)

Definition: Distance from a lens (or mirror) to its focal point. Shorter f = stronger converging. Convex: real F; Concave: virtual F.

Example: A magnifying glass with $f = 10$ cm has its focal point 10 cm from the lens. The Sun's rays focused to a tiny spot at this point can ignite paper.

Cognates: *Sp. distancia focal · Pt. distância focal · HC distans fokal*

Watch out: The 'f' in focal length is NOT the same as the 'f' for frequency. Context.

VOCAB · REAL VS. VIRTUAL IMAGE

Definition: REAL image: light rays actually converge to the image point. CAN be caught on a screen. VIRTUAL image: rays only APPEAR to converge (after tracing back). Cannot be caught on a screen.

Example: Movie projector → REAL image on the screen. Magnifying glass close to small print → VIRTUAL magnified image (you see it through the lens).

Cognates: *Sp. imagen real/virtual · Pt. imagem real/virtual*

Watch out: Real ↔ catchable on a screen. Virtual ↔ only seen by an eye/camera looking through the lens.

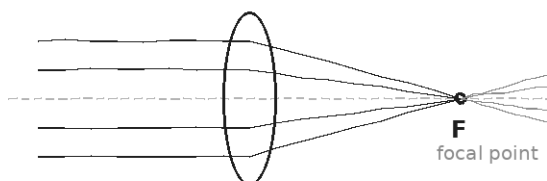
DOODLE ZONE

TOP: parallel rays hit a convex lens — converge to focal point F . BOTTOM: same rays hit a concave lens — diverge outward; trace back (dashed) to a virtual F . Which lens is in a magnifying glass? In nearsighted glasses?

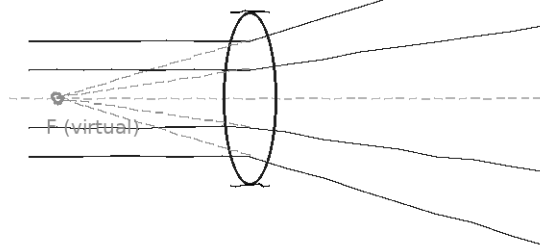
LENSES · convex CONVERGES. Concave DIVERGES.

Parallel rays come in. Lens bends them through (or away from) a focal point.

A — CONVEX LENS (converges)



B — CONCAVE LENS (diverges)



Which one is used in a magnifying glass? ____ In a nearsighted person's glasses? ____

WORKED EXAMPLE — CLASSIFY EACH APPLICATION

For each application, identify the lens type (convex/concave) and image type (real/virtual): (a) magnifying glass; (b) camera; (c) movie projector; (d) nearsighted glasses; (e) door peephole.

GIVEN	—
EQUATION	(qualitative reasoning — no equation)
WORK	(a) magnifying glass: CONVEX, VIRTUAL (close to object) (b) camera: CONVEX, REAL (image on sensor) (c) movie projector: CONVEX, REAL (image on screen) (d) nearsighted glasses: CONCAVE, VIRTUAL (diverges focus) (e) peephole: CONCAVE, VIRTUAL (wide-angle view)
ANSWER	3 convex (2 real, 1 virtual), 2 concave (both virtual).

YOUR TURN — BURN PAPER WITH THE SUN

You hold a magnifying glass (convex, $f = 10$ cm) above a piece of dry paper on a sunny day. At what distance above the paper do you hold the lens to focus the Sun's image and ignite the paper?

GIVEN	
EQUATION	
WORK	
ANSWER	



That last one is real — kids burn ants with magnifying glasses for a reason. The focused Sun's image at the focal point has roughly $1000\times$ the intensity of unfocused sunlight. Inverse-square-law (Day 11) in reverse: small area $\rightarrow$ concentrated power.

Fill in the frame:

A _____ lens brings parallel rays to a real focal point — it CONVERGES light. A _____ lens spreads them out, producing only virtual images. The KIND of image depends on the lens shape and the object's _____.

EXIT TICKET — DAY 14

Why does looking through a magnifying glass make small print bigger? In one sentence, say what kind of lens it is + what kind of image is formed.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can distinguish convex vs. concave lenses by shape and behavior.	☐	☐	☐	☐
I can identify real vs. virtual images.	☐	☐	☐	☐
I can match common applications to lens type.	☐	☐	☐	☐

Day 15 — ★ RADAR — How NASA Measured 2026-XJ (The Reveal)

Today's Target

I can trace the FULL radar measurement chain: send a pulse, time the round trip for DISTANCE ($d = c \cdot \Delta t / 2$), measure the frequency SHIFT for VELOCITY ($v = (\Delta f / f) \cdot c$), and estimate SIZE from signal strength.

ASTEROID THREAD

What we know about 2026-XJ today: For five units we've USED 2026-XJ numbers. Today we see WHERE THEY CAME FROM.

How today connects: Every Unit 6 idea is required: $v = f\lambda$ (Day 3), reflection (Day 4), Doppler (Day 11), EM waves (Day 12), mirrors (Day 13). Today they combine into the radar measurement chain.

★ THE REVEAL · WHERE THE NUMBERS CAME FROM

Since Unit 1 you have used these 2026-XJ numbers: distance from Earth, closing velocity, mass, trajectory, impact KE, damage radii. Every one had to be MEASURED. The instrument is RADAR — specifically NASA's Goldstone (California) and Arecibo-era (Puerto Rico) planetary radar systems. Modern planetary radar transmits 2.4 GHz radio pulses at very high power, listens for the echo from the asteroid, and extracts three pieces of information from the return:

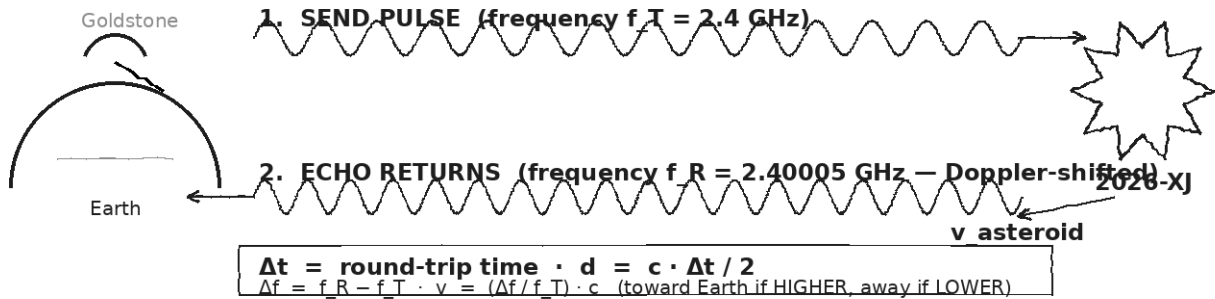
- **DISTANCE:** from the round-trip TIME Δt : $d = c \cdot \Delta t / 2$. At $c = 3 \times 10^8$ m/s, a $\Delta t = 2.5$ s means the asteroid is 375 000 km away.
- **VELOCITY (line-of-sight):** from the DOPPLER SHIFT Δf : $v = (\Delta f / f) \cdot c$. A 5 kHz shift on a 2.4 GHz carrier means $v \approx 312$ m/s closing (or receding).
- **SIZE (estimate):** from signal STRENGTH vs. distance: a stronger echo means a larger radar cross-section. Combined with an assumed albedo + density, size and mass estimates fall out — with uncertainty.

DOODLE ZONE

The unit's centerpiece. Trace the full chain. Mark Δt (round-trip time). Compare outgoing wave (f_T) to returning wave (f_R) — does it look compressed or stretched? What does that tell you about asteroid motion?

★ RADAR · How NASA actually measured 2026-XJ

Send → bounce → receive. Time tells distance. Frequency SHIFT tells velocity.



STEP 1 — DISTANCE FROM ROUND-TRIP TIME

A radar pulse is sent toward 2026-XJ. The echo returns $\Delta t = 2.5 \text{ s}$ later. How far away is the asteroid?

GIVEN	$\Delta t = 2.5 \text{ s}$ · $c = 3.00 \times 10^8 \text{ m/s}$
EQUATION	$d = c \cdot \Delta t / 2$ (round trip → one way)
WORK	$d = (3.00 \times 10^8) \cdot 2.5 / 2$ $d = 7.5 \times 10^8 / 2$ $d = 3.75 \times 10^8 \text{ m}$ $d = 375\,000 \text{ km}$
ANSWER	$d \approx 375\,000 \text{ km}$ — about the distance to the Moon. Real-world precision: $\pm 100 \text{ m}$.

STEP 2 — VELOCITY FROM THE DOPPLER SHIFT

The transmitted radar pulse is at $f_T = 2.4 \text{ GHz}$. The echo returns at $f_R = 2.40005 \text{ GHz}$ (5 kHz HIGHER). What's the asteroid's line-of-sight velocity? Is it approaching or receding?

GIVEN	$f_T = 2.4 \times 10^9 \text{ Hz} \cdot f_R = 2.400005 \times 10^9 \text{ Hz} \cdot \Delta f = 5\,000 \text{ Hz} \cdot c = 3 \times 10^8 \text{ m/s}$
EQUATION	$v = (\Delta f / f_T) \cdot c$ (HIGHER echo → APPROACHING)
WORK	$v = (5000 / 2.4 \times 10^9) \cdot 3 \times 10^8$ $v = (2.08 \times 10^{-6}) \cdot 3 \times 10^8$ $v \approx 625 \text{ m/s}$
ANSWER	$v \approx 625 \text{ m/s}$, APPROACHING (echo frequency was higher than transmitted). Modern systems detect Δf down to 0.01 Hz — ~mm/s velocity precision.

STEP 3 — SIZE ESTIMATE FROM SIGNAL STRENGTH

The echo's STRENGTH (energy per square meter back at Earth) depends on (a) how far the wave traveled ($1/r^4$ — out AND back), and (b) the asteroid's RADAR CROSS-SECTION (effective area). Why does the signal weaken as $1/r^4$, not just $1/r^2$?

GIVEN	Wave travels: Earth → asteroid → Earth. Two legs, each inverse-square.
EQUATION	$I \propto 1 / (r^2 \times r^2) = 1 / r^4$
WORK	Outgoing wave to asteroid: I drops as $1/r^2$ (Day 11). After reflecting, echo wave back to Earth: ANOTHER $1/r^2$ drop. Total: $1/r^4$. So radar at TWICE the distance is $16\times$ weaker.
ANSWER	$1/r^4$ is brutal — that's why planetary radar transmits MEGAWATT pulses. Larger asteroid → larger cross-section → stronger echo → size estimate.

★	The numbers we've been using are REAL outputs of this chain. The 2026-XJ mass estimate has the largest uncertainty (depends on assumed albedo + density). The velocity is precise to mm/s. The distance is precise to meters. That's WHY the impact KE has uncertainty — not because the physics is wrong, but because the measurement is bounded by what radar can resolve.
---	--

Fill in the frame:

Radar measures asteroid distance from

_____ (the round-trip time times

c , halved). Radar measures velocity from

_____ (the Doppler frequency

shift). Higher echo frequency → asteroid is

_____ ; lower → receding. Mass
uncertainty comes from the fact that signal STRENGTH only gives a
_____ estimate of size.

EXIT TICKET — DAY 15

Why does NASA need MULTIPLE radar observations of the same asteroid (over weeks or months) to predict its orbit precisely?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can apply $d = c \cdot \Delta t / 2$ to compute asteroid distance from radar.	☐	☐	☐	☐
I can apply $v = (\Delta f / f) \cdot c$ to compute Doppler-derived velocity.	☐	☐	☐	☐
I can explain why radar power falls as $1/r^4$ for an echo.	☐	☐	☐	☐
I can identify which 2026-XJ quantity has the biggest measurement uncertainty.	☐	☐	☐	☐

Day 16 — Meteor Color · Wien's Law: Hotter Objects Glow at Shorter λ

Today's Target

I can apply Wien's law qualitatively (and the constant $2.9 \times 10^{-3} \text{ m}\cdot\text{K}$ quantitatively if stretched) to predict the COLOR of a thermally radiating object — meteor, star, stovetop — from its TEMPERATURE.

ASTEROID THREAD

What we know about 2026-XJ today: A meteor's atmospheric-entry glow (Unit 5 Day 10) heats it to 5000–10 000 K. It glows yellow-white, sometimes blue-white.

How today connects: Wien's law explains WHY. Same rule that classifies stars by color. Both 2026-XJ's entry glow AND the impact fireball follow this law.

VOCAB · BLACKBODY RADIATION

Definition: The EM radiation emitted by ALL warm objects, with a spectrum determined entirely by temperature. Hotter object = higher peak frequency.

Example: A red-hot iron, a glowing meteor, the Sun, the Cosmic Microwave Background — all blackbodies at different T.

Cognates: *Sp. radiación de cuerpo negro · Pt. radiação de corpo negro*

Watch out: A blackbody doesn't have to look black. It's an idealization. Real objects approximate it well at high T.

VOCAB · WIEN'S LAW

Definition: The peak wavelength of a blackbody's emission is inversely proportional to its temperature: $\lambda_{\text{max}} \cdot T = 2.9 \times 10^{-3} \text{ m}\cdot\text{K}$. Higher T $\rightarrow$ shorter λ_{max} (toward blue).

Example: The Sun at $T \approx 6000 \text{ K} \rightarrow \lambda_{\text{max}} \approx 500 \text{ nm}$ (yellow-green). A red-hot iron at $1000 \text{ K} \rightarrow \lambda_{\text{max}} \approx 3 \mu\text{m}$ (mostly infrared; only the longest visible reds are seen).

Cognates: *Sp. ley de Wien · Pt. lei de Wien*

Watch out: At very high T (10 000 K+), the peak shifts into UV — but visible RED + blue together look BLUE-WHITE to us.

VOCAB · COLOR TEMPERATURE

Definition: A way of expressing the COLOR of a thermally radiating object as a temperature (in Kelvin). Used in photography, astronomy, lighting design.

Example: Incandescent bulb: 2700 K (warm yellow). Sunlight: 5500-6500 K (white). Cloudy sky: 10 000 K (blue tint).

Cognates: *Sp. temperatura de color · Pt. temperatura de cor*

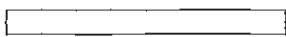
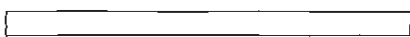
Watch out: Higher color temperature looks BLUER. Counterintuitive: "warm" light is LOWER T; "cool" light is HIGHER T.

DOODLE ZONE

Order from coolest to hottest: red-hot iron, yellow flame, the Sun, atmospheric meteor, blue star. Each step UP in temperature shifts peak emission toward SHORTER wavelengths.

WIEN'S LAW · hotter objects glow at SHORTER wavelengths (higher f).

Color tells temperature. Stars, stovetops, atmospheric-entry meteors — same rule.

RED-HOT IRON	1000 K	red	
YELLOW FLAME	1700 K	orange	
INCANDESCENT BULB	3000 K	yellow	
THE SUN	6000 K	yellow-white	
ATMOSPHERIC METEOR	5000-10 000 K	YELLOW-WHITE	
VEGA (bright star)	10 000 K	white-blue	
RIGEL (hottest stars)	30 000 K	BLUE	

Wien: $\lambda_{\text{max}} \cdot T = 2.9 \times 10^{-3} \text{ m} \cdot \text{K}$ · meteor at 5000 K $\rightarrow \lambda_{\text{max}} \approx 580 \text{ nm} = \text{YELLOW}$

WORKED EXAMPLE — METEOR PEAK WAVELENGTH

A meteor at atmospheric entry is heated to $T \approx 5000 \text{ K}$. Compute the peak emission wavelength using Wien's law: $\lambda_{\text{max}} \cdot T = 2.9 \times 10^{-3} \text{ m}\cdot\text{K}$. What color does this correspond to?

GIVEN	$T = 5000 \text{ K} \cdot b = 2.9 \times 10^{-3} \text{ m}\cdot\text{K}$
EQUATION	$\lambda_{\text{max}} = b / T$
WORK	$\lambda_{\text{max}} = (2.9 \times 10^{-3}) / 5000$ $\lambda_{\text{max}} = 5.8 \times 10^{-7} \text{ m}$ $\lambda_{\text{max}} = 580 \text{ nm}$
ANSWER	$\lambda_{\text{max}} \approx 580 \text{ nm}$ — YELLOW. (The peak's in yellow; the wings reach red + green, making the meteor look YELLOW-WHITE.)

WORKED EXAMPLE — EXTREME ENTRY SPEED

A faster meteor (or an asteroid at high entry speed) can heat to $T \approx 10\,000 \text{ K}$. What's the peak λ ? What color?

GIVEN	$T = 10\,000 \text{ K} \cdot b = 2.9 \times 10^{-3} \text{ m}\cdot\text{K}$
EQUATION	$\lambda_{\text{max}} = b / T$
WORK	$\lambda_{\text{max}} = (2.9 \times 10^{-3}) / 10\,000$ $\lambda_{\text{max}} = 2.9 \times 10^{-7} \text{ m}$ $\lambda_{\text{max}} = 290 \text{ nm}$
ANSWER	$\lambda_{\text{max}} \approx 290 \text{ nm}$ — UV. The visible part looks BLUE-WHITE (heavily weighted toward blue + violet).

YOUR TURN — CLASSIFY BY COLOR

A red-hot iron glows at $T \approx 1000 \text{ K}$. What's λ_{max} ? Why does it look RED to us (despite λ_{max} being beyond visible)?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	Astronomers measure star COLOR to estimate surface temperature. Cool stars (M-class, 3000 K) are RED. Sun-like stars (G-class, 6000 K) are YELLOW. Hot stars (O-class, 30 000 K) are BLUE. Same physics as the meteor glow.
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Fill in the frame:

Hotter object → peak emission at

_____ wavelength. The Sun

(6000 K) peaks at _____ nm. A

10 000 K meteor peaks at _____

nm — in the _____ band. Visible

color combines the WHOLE spectrum, weighted by intensity.

EXIT TICKET — DAY 16

A meteor entering at extreme speed could heat to 10 000 K. What color would it glow? How does this compare to a typical meteor (5000 K)? Give a one-sentence comparison.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can apply Wien's law to predict peak wavelength from temperature.	☐	☐	☐	☐
I can connect color to temperature (red cool, yellow warm, blue hot).	☐	☐	☐	☐
I can explain the meteor's yellow-white glow.	☐	☐	☐	☐

Day 17 — Seismic + Tsunami · Mechanical-Wave

Damage at Continental Scale

Today's Target

I can predict the arrival time of seismic P-waves and S-waves at distant locations, and apply $v = \sqrt{g \cdot d}$ to estimate tsunami speed + arrival time at a distant coastline.

ASTEROID THREAD

What we know about 2026-XJ today: 2026-XJ has a 70% chance of impacting OCEAN. If it does, the dominant long-range damage mechanism shifts to TSUNAMI. If it hits land, SEISMIC waves register globally.

How today connects: Today's payoff: both kinds of mechanical waves carry impact damage far beyond the immediate strike zone. Wave physics scales up to planetary geometry.

VOCAB · SEISMIC P-WAVE (PRIMARY)

Definition: A LONGITUDINAL mechanical wave traveling through Earth's crust + mantle. Compressions and rarefactions of rock. ≈ 6 km/s in mantle. FIRST to arrive at a seismograph (hence 'P').

Example: An earthquake or large impact launches P-waves. Diameter of Earth (~ 12700 km) / 6 km/s ≈ 35 min — P-waves traverse the whole planet in about half an hour.

Cognates: *Sp. onda P · Pt. onda P · HC vag P*

Watch out: P-waves are LONGITUDINAL like sound. They can travel through solid AND liquid. They're the only seismic wave that crosses Earth's liquid outer core.

VOCAB · SEISMIC S-WAVE (SECONDARY)

Definition: A TRANSVERSE mechanical wave through Earth's crust + mantle. Side-to-side oscillation of rock. ≈ 4 km/s in mantle. ARRIVES SECOND.

Example: Follows P-wave by minutes. The TIME GAP between P and S arrival is how seismologists triangulate the source distance.

Cognates: *Sp. onda S · Pt. onda S · HC vag S*

Watch out: S-waves can NOT travel through liquids. That's how we know Earth's outer core is liquid — S-waves don't make it through. P-waves do.

VOCAB · TSUNAMI

Definition: An ocean-surface gravity wave triggered by a sudden displacement of water (impact, undersea earthquake, landslide). Speed $v = \sqrt{g \cdot d}$, where d is ocean depth.

Example: A 2026-XJ ocean impact at 4 km depth generates a tsunami moving at $v = \sqrt{9.8 \cdot 4000} \approx 200$ m/s ≈ 700 km/h. Reaches a coast 1500 km away in about 2 hours.

Cognates: *Sp. tsunami · Pt. tsunami · HC tsunami*

Watch out: In DEEP water, a tsunami is only ~ 1 m tall but moves fast. As it reaches SHALLOW water, it slows AND grows tall (energy conservation). Coast is where damage happens.

DOODLE ZONE

Cross-section of the ocean. Tsunami in DEEP water is fast + low. As it reaches the SHALLOW coast, it slows AND amplifies. Where does the most energy build up? What happens at the shoreline?

TSUNAMI · ocean-surface gravity wave · $v = \sqrt{g \cdot d}$

Deep ocean: FAST + LOW amplitude. Shallow coast: SLOW + TALL.

DEEP OCEAN ($d \approx 4$ km)

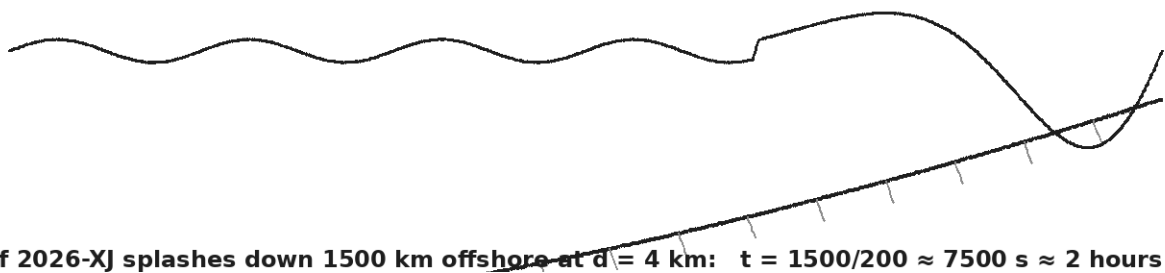
$v = \sqrt{g \cdot d} \approx \sqrt{9.8 \cdot 4000} \approx 200$ m/s ≈ 700 km/h

Wave height: only ~ 1 m — hard to see from a boat!

SHALLOW COAST ($d \approx 10$ m)

v drops to ~ 10 m/s — but ENERGY conserved.

Wave grows to 10+ m tall: catastrophic.



If 2026-XJ splashes down 1500 km offshore at $d = 4$ km: $t = 1500/200 \approx 7500$ s ≈ 2 hours to coast

WORKED EXAMPLE — P-WAVE ARRIVAL ACROSS EARTH

An asteroid impact in Africa launches seismic waves. How long until the P-waves are detected by a seismograph in California, about 13 000 km away through Earth?

GIVEN	$d = 13\,000\text{ km} = 1.3 \times 10^7\text{ m}$ · $v_P = 6\text{ km/s} = 6000\text{ m/s}$
EQUATION	$t = d / v$
WORK	$t = 1.3 \times 10^7 / 6000$ $t = 2167\text{ s}$ $t \approx 36\text{ minutes}$
ANSWER	$\approx 36\text{ min}$ after impact. S-waves arrive $\sim 18\text{ minutes}$ later (slower at 4 km/s).

WORKED EXAMPLE — TSUNAMI SPEED IN DEEP OCEAN

An ocean impact occurs at $d = 4\text{ km}$ depth. What's the tsunami's deep-ocean speed?

GIVEN	$g = 9.8\text{ m/s}^2$ · $d = 4000\text{ m}$
EQUATION	$v = \sqrt{g \cdot d}$
WORK	$v = \sqrt{9.8 \cdot 4000}$ $v = \sqrt{39\,200}$ $v \approx 198\text{ m/s}$ $v \approx 200\text{ m/s} \approx 700\text{ km/h}$
ANSWER	$\approx 200\text{ m/s} \approx 700\text{ km/h}$. As fast as a jet airliner.

WORKED EXAMPLE — ARRIVAL TIME ON A DISTANT COAST

Continuing the example above: how long until that tsunami reaches a coastline 1500 km away?

GIVEN	$d = 1.5 \times 10^6 \text{ m} \cdot v = 200 \text{ m/s}$
EQUATION	$t = d / v$
WORK	$t = 1.5 \times 10^6 / 200$ $t = 7500 \text{ s}$ $t \approx 125 \text{ min}$ $t \approx 2.1 \text{ hours}$
ANSWER	≈ 2 hours warning before impact at the coast. That window is why tsunami early-warning networks (DART buoys + seismic detection) exist.

YOUR TURN — S-WAVE ARRIVAL

After the same African impact, how long until the S-waves arrive in California (13 000 km away, $v_S = 4 \text{ km/s}$)? Compute the GAP between P-wave and S-wave arrival.

GIVEN	
EQUATION	
WORK	
ANSWER	

WAVE DAMAGE-BY-DISTANCE · 2026-XJ OCEAN IMPACT SCENARIO

► Atmospheric shockwave	minutes	few hundred km	broken windows; flattened forest
► Thermal pulse (Unit 5)	seconds	~800 km	wildfire ignition radius
► Seismic P-wave	~35 min	global (~12 700 km)	registers worldwide
► Seismic S-wave	~50 min	via mantle (no liquid core)	global on land
► Tsunami	~2 hours	thousands of km	catastrophic at the coast



If 2026-XJ splashes down 1500 km from a coast, the THERMAL pulse arrives in seconds, the SEISMIC arrival in ~35 min, but the TSUNAMI takes ~2 hours. That's the warning window. Whether the warning saves lives depends on how fast it's relayed.

Fill in the frame:

Seismic P-waves are _____

(longitudinal/transverse) and travel

~ _____ km/s. S-waves are

_____ and travel

~ _____ km/s. A tsunami moves

at $v = \sqrt{(\text{_____})}$, so deeper ocean

→ _____ tsunami.

EXIT TICKET — DAY 17

Why is a tsunami much more dangerous on the shoreline than in deep ocean? Answer in two sentences using the phrase 'energy conservation'.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can compute P-wave + S-wave arrival times at distant locations.	☐	☐	☐	☐
I can apply $v = \sqrt{g \cdot d}$ to tsunami speed.	☐	☐	☐	☐
I can predict tsunami arrival time on a distant coast.	☐	☐	☐	☐
I can explain why a tsunami slows AND amplifies toward shore.	☐	☐	☐	☐

Day 18 — Unit 6 Transfer Task — Assessment Day

ASTEROID THREAD

What we know about 2026-XJ today: Every Unit 6 tool: $v = f\lambda$, reflection, refraction, diffraction, interference, Doppler, dB, EM spectrum, mirrors, lenses, Wien's law, tsunami $v = \sqrt{g \cdot d}$, seismic speeds.

How today connects: Problem 5 is the load-bearing R4 prompt: given a radar return + Doppler shift, compute distance + velocity, then predict tsunami arrival on a coast. The whole unit in one problem.

TRANSFER TASK — UNIT 6 · DAY 18

Today you receive the Unit 6 Transfer Task on a separate sheet. It is 5 multi-part problems. You will work independently for ~40 minutes.

- Problem 1 — WAVE EQUATION ACROSS MEDIA. A 1000 Hz sound moves from air ($v = 343$ m/s) into water ($v = 1480$ m/s). Compute λ in each medium. Explain what stays the same and what changes (and why).
- Problem 2 — REFLECTION. A flat mirror is 2.0 m from a 1.6-m-tall person. Locate the image and compute its height. State whether real or virtual. Sketch a 2-ray diagram.
- Problem 3 — DOPPLER. A 1000 Hz ambulance siren moves at 25 m/s toward a stationary listener. What frequency is heard? What about the listener BEHIND the ambulance (away)? Use $f' = f \cdot v / (v \mp v_{\text{source}})$.
- Problem 4 — EM SPECTRUM. A radio station broadcasts at $f = 540$ kHz. Compute λ . Which EM band is this? (Use $c = 3 \times 10^8$ m/s.) Is this AM or FM?
- Problem 5 — ★ ASTEROID SCENARIO. Radar pulse to 2026-XJ: transmitted $f_T = 2.4$ GHz, round-trip time $\Delta t = 2.4$ s, echo received at $f_R = 2.40010$ GHz (10 kHz higher). (a) Compute distance. (b) Compute approach velocity. (c) If 2026-XJ impacts the ocean 1500 km from a coast ($d_{\text{ocean}} = 4$ km), how long until the tsunami arrives at the coast?

DOODLE ZONE

Five problems, five physics ideas. Match each problem to its Unit 6 idea *BEFORE* you begin.
Problem 5 is the ★ — it integrates radar + Doppler + tsunamis.

UNIT 6 TRANSFER TASK · 5 multi-part problems

Match each problem to its Unit 6 idea before you start.

(1) $v = f\lambda$ across media

sound from air → water

(2) reflection

plane mirror image location

(3) Doppler shift

moving sound source

(4) EM spectrum

identify band + compute λ

(5) ★ asteroid scenario

radar $\Delta t + \Delta f \rightarrow$ tsunami arrival

HOW THIS WILL BE SCORED — 4 DIMENSIONS, EACH 0-4

Dimension	What a strong (3-4) response shows
Science	Correct physics: vectors, kinematics, units.
Reasoning	Sound method; uncertainty explained, not just stated.
Communication	Work shown, units labeled, organized, readable.
Transfer-to-Asteroid	Honest public-facing connection to 2026-XJ.

MY WORK — DAY 18 (USE ADDITIONAL PAPER AS NEEDED)

AFTER THE TASK — FINAL UNIT MARZANO (RATE EACH UNIT 6 TARGET)

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
K1 — Waves carry energy/info without carrying matter.	☐	☐	☐	☐
K2 — Wave properties: f , λ , A , T (and $f = 1/T$).	☐	☐	☐	☐
K3 — $v = f \cdot \lambda$ across media.	☐	☐	☐	☐
K4 — Reflection, refraction, diffraction qualitatively.	☐	☐	☐	☐
K5 — Interference + standing waves.	☐	☐	☐	☐
K6 — Sound as longitudinal wave; speed in media.	☐	☐	☐	☐
K7 — Intensity, inverse-square, decibel scale.	☐	☐	☐	☐
K8 — Doppler effect.	☐	☐	☐	☐
K9 — EM spectrum and $c = f\lambda$.	☐	☐	☐	☐
K10 — Mirrors + lenses qualitatively.	☐	☐	☐	☐
K11 — Wien's law (color $\leftrightarrow$ temperature).	☐	☐	☐	☐
K12 — Seismic + tsunami wave physics.	☐	☐	☐	☐
S1 — Compute v , f , or λ given the other two.	☐	☐	☐	☐
S2 — Apply Doppler to compute observed frequency.	☐	☐	☐	☐
S4 — Use Vernier to measure v_{sound} by echo.	☐	☐	☐	☐
R4 — Apply EM-wave framework to radar measurement of 2026-XJ.	☐	☐	☐	☐

Equation Reference — Unit 6

Keep this page open while you work. Every equation below is on the MCAS Equation Sheet — you do NOT need to memorize them. You DO need to recognize when each one applies.

Concept	Equation	When to use it
Frequency ↔ period	$f = 1 / T$	Cycles per second is the reciprocal of seconds per cycle. $\text{Hz} = 1/\text{s}$.
Wave equation	$v = f \cdot \lambda$	Speed = frequency × wavelength. THE central equation of Unit 6. Speed is fixed by medium; the source sets f ; λ adjusts to fit.
Law of reflection	$\theta_i = \theta_r$	Angle of incidence = angle of reflection — both measured from the NORMAL (perpendicular to the surface).
Refraction (qualitative)	slower medium → bend TOWARD normal	Faster medium → bend AWAY from normal. Larger speed change = larger bend. (Snell's law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$ — see if interested.)
Diffraction (qualitative)	smaller opening (vs. λ) → more spreading	Long λ bends around everyday objects (sound, radio). Short λ does not (light, X-ray).
Sound intensity (inverse square)	$I \propto 1 / r^2$	Intensity drops as the square of distance from a point source. Twice as far = 1/4 the intensity.
Decibel scale	$\text{dB} = 10 \cdot \log_{10}(I / I_0)$	$I_0 = 10^{-12} \text{ W/m}^2$ (threshold of hearing). Every +10 dB = ×10 in intensity. 0 dB threshold, 60 dB conversation, 120 dB pain.
Doppler shift (sound)	$f' = f \cdot v_{\text{sound}} / (v_{\text{sound}} - v_{\text{source}})$	Moving TOWARD listener: use minus → higher f . Moving AWAY: use plus → lower f . Same idea applies to radar.
Radar distance	$d = c \cdot \Delta t / 2$	Round-trip time × c , then halve. Day 15: 2026-XJ at $\Delta t = 2.5 \text{ s} \rightarrow d = 375,000 \text{ km}$.
Radar Doppler velocity	$v = (\Delta f / f) \cdot c$	Echo frequency shift divided by transmit frequency, times c . Higher echo f = approaching; lower f = receding.
Wien's law (qualitative)	$\lambda_{\text{max}} \cdot T = 2.9 \times 10^{-3} \text{ m} \cdot \text{K}$	Hotter object → peak emission at SHORTER wavelength. Cool star red, Sun yellow-white, hot star blue.

Tsunami speed	$v = \sqrt{g \cdot d}$	Ocean-surface gravity wave. d = ocean depth in meters. At $d = 4$ km, $v \approx 200$ m/s ≈ 700 km/h.
Standing wave on a string	$f_n = n \cdot v / (2L)$	Both ends fixed. $n = 1$ fundamental, $n = 2$ second harmonic, etc. Determines pitch of guitar strings + organ pipes.

ELECTROMAGNETIC SPECTRUM — BANDS AND USES

Band	λ (vacuum)	Use
Radio waves	> 1 m	Broadcasting, cell, planetary radar (Day 15).
Microwaves	1 m to 1 mm	Microwave oven (2.4 GHz), radar.
Infrared (IR)	1 mm to 700 nm	Heat, night vision, TV remote.
Visible light	700 to 400 nm	ROYGBIV — what our eyes detect.
Ultraviolet (UV)	400 to 10 nm	Sunburn, sterilization.
X-rays	10 to 0.01 nm	Medical imaging.
Gamma rays	< 0.01 nm	Nuclear processes, cosmic events.

WAVE SPEEDS — REFERENCE VALUES

Wave + medium	Speed	Note
Light / radio in vacuum	$c = 3.00 \times 10^8 \text{ m/s}$	Universal speed limit.
Sound in air (20 °C)	$v \approx 343 \text{ m/s}$	Speeds up about 0.6 m/s per °C warmer.
Sound in water	$v \approx 1480 \text{ m/s}$	About 4× faster than in air.
Sound in steel	$v \approx 5960 \text{ m/s}$	Solids transmit sound fastest.
Seismic P-wave (mantle)	$v \approx 6 \text{ km/s}$	Longitudinal — first to arrive.
Seismic S-wave (mantle)	$v \approx 4 \text{ km/s}$	Transverse — arrives second; can't travel through liquid outer core.
Tsunami (deep ocean, 4 km)	$v \approx 200 \text{ m/s} \approx 700 \text{ km/h}$	$v = \sqrt{g \cdot d}$ — slows AND amplifies near shore.

DECIBEL REFERENCE — COMMON SOUND LEVELS

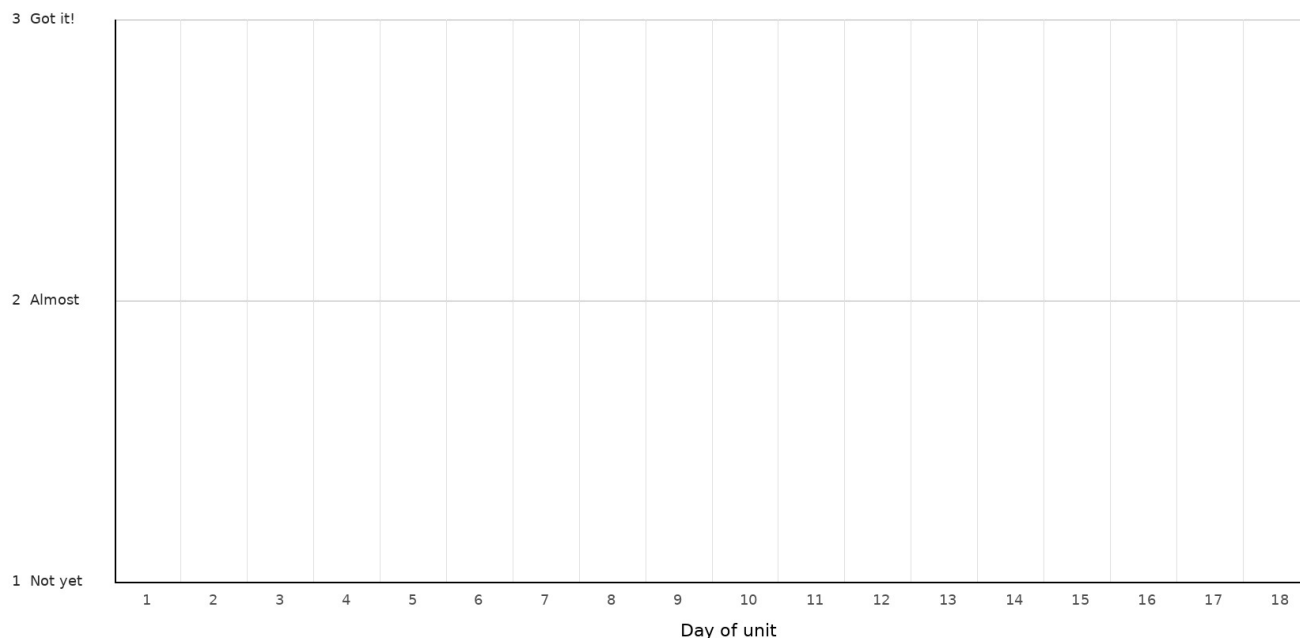
Level	Source
0 dB	Threshold of hearing
30 dB	Quiet library, whisper
60 dB	Normal conversation
85 dB	Heavy traffic — hearing damage with prolonged exposure
100 dB	Loud rock concert, chainsaw
120 dB	Pain threshold, thunderclap
194 dB	Theoretical max in air (atmospheric shockwave)

Constants and reference quantities:

$c = 3.00 \times 10^8 \text{ m/s}$ · $v_{\text{sound (air, 20 °C)}} \approx 343 \text{ m/s}$ · $I_0 = 10^{-12} \text{ W/m}^2$ · $g = 9.8 \text{ m/s}^2$ · Wien's constant $b = 2.9 \times 10^{-3} \text{ m} \cdot \text{K}$ · ROYGBIV: 700 → 400 nm · Earth diameter $\approx 12\,700 \text{ km}$

Mastery Growth Chart — Unit 6

Each day, transfer your Marzano 1-2-3 rating onto the chart. Use a different mark or color for each target you track. Unit 6 has ~13 targets across 18 days — track at least your top three.



LEGEND — WHICH MARK / COLOR = WHICH TARGET

Mark: _____	Target: _____
Mark: _____	Target: _____
Mark: _____	Target: _____
Mark: _____	Target: _____

NOTES — END-OF-UNIT REFLECTION